

FUNDAMENTALS OF CONVECTION

So far, we have considered *conduction*, which is the mechanism of heat transfer through a solid or a quiescent fluid. We now consider *convection*, which is the mechanism of heat transfer through a fluid in the presence of bulk fluid motion.

Convection is classified as *natural* (or *free*) and *forced convection*, depending on how the fluid motion is initiated. In forced convection, the fluid is forced to flow over a surface or in a pipe by external means such as a pump or a fan. In natural convection, any fluid motion is caused by natural means such as the buoyancy effect, which manifests itself as the rise of warmer fluid and the fall of the cooler fluid. Convection is also classified as *external* and *internal*, depending on whether the fluid is forced to flow over a surface or in a pipe.

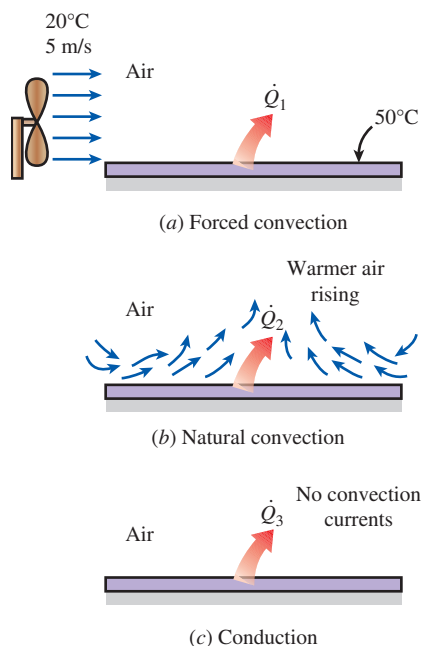
We start this chapter with a general physical description of the convection mechanism. We then discuss the *velocity* and *thermal boundary layers* and *laminar and turbulent flows*. We continue with the discussion of the dimensionless *Reynolds*, *Prandtl*, and *Nusselt numbers* and their physical significance. Next we derive the *convection equations* on the basis of mass, momentum, and energy conservation, and we obtain solutions for *flow over a flat plate*. We then nondimensionalize the convection equations and obtain functional forms of friction and convection coefficients. Finally, we present analogies between momentum and heat transfer.



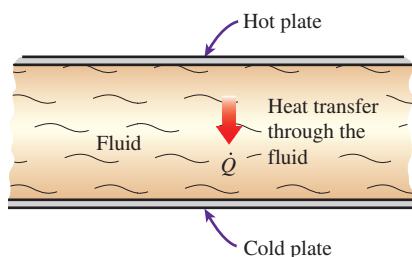
OBJECTIVES

When you finish studying this chapter, you should be able to:

- Understand the physical mechanism of convection and its classification,
- Visualize the development of velocity and thermal boundary layers during flow over surfaces,
- Gain a working knowledge of the dimensionless Reynolds, Prandtl, and Nusselt numbers,
- Distinguish between laminar and turbulent flows, and gain an understanding of the mechanisms of momentum and heat transfer in turbulent flow,
- Derive the differential equations that govern convection on the basis of mass, momentum, and energy balances, and solve these equations for some simple cases such as laminar flow over a flat plate,
- Nondimensionalize the convection equations and obtain the functional forms of friction and heat transfer coefficients, and
- Use analogies between momentum and heat transfer, and determine heat transfer coefficient from your knowledge of friction coefficient.

**FIGURE 6-1**

Heat transfer from a hot surface to the surrounding fluid by convection and conduction.

**FIGURE 6-2**

Heat transfer through a fluid sandwiched between two parallel plates.

6-1 ■ PHYSICAL MECHANISM OF CONVECTION

We mentioned in Chap. 1 that there are three basic mechanisms of heat transfer: conduction, convection, and radiation. Conduction and convection are similar in that both mechanisms require the presence of a material medium. But they are different in that convection requires the presence of fluid motion.

Heat transfer through a solid is always by conduction, since the molecules of a solid remain at relatively fixed positions. Heat transfer through a liquid or gas, however, can be by conduction or convection, depending on the presence of any bulk fluid motion. Heat transfer through a fluid is by convection in the presence of bulk fluid motion and by conduction in the absence of it. Therefore, conduction in a fluid can be viewed as the limiting case of convection, corresponding to the case of quiescent fluid (Fig. 6-1).

Convection heat transfer is complicated by the fact that it involves fluid motion as well as heat conduction. The fluid motion enhances heat transfer, since it brings warmer and cooler chunks of fluid into contact, initiating higher rates of conduction at a greater number of sites in a fluid. Therefore, the rate of heat transfer through a fluid is much higher by convection than it is by conduction. In fact, the higher the fluid velocity, the higher the rate of heat transfer.

To clarify this point further, consider steady heat transfer through a fluid contained between two parallel plates maintained at different temperatures, as shown in Fig. 6-2. The temperatures of the fluid and the plate are the same at the points of contact because of the continuity of temperature. Assuming no fluid motion, the energy of the hotter fluid molecules near the hot plate is transferred to the adjacent cooler fluid molecules. This energy is then transferred to the next layer of the cooler fluid molecules. This energy is then transferred to the next layer of the cooler fluid, and so on, until it is finally transferred to the other plate. This is what happens during conduction through a fluid. Now let us use a syringe to draw some fluid near the hot plate and inject it next to the cold plate repeatedly. You can imagine that this will speed up the heat transfer process considerably, since some energy is carried to the other side as a result of fluid motion.

Consider the cooling of a hot block with a fan blowing air over its top surface. We know that heat is transferred from the hot block to the surrounding cooler air, and the block eventually cools. We also know that the block cools faster if the fan is switched to a higher speed. Replacing air with water enhances the convection heat transfer even more.

Experience shows that convection heat transfer strongly depends on the fluid properties *dynamic viscosity* μ , *thermal conductivity* k , *density* ρ , and *specific heat* c_p , as well as the *fluid velocity* V . It also depends on the *geometry* and the *roughness* of the solid surface, in addition to the *type of fluid flow* (such as being streamlined or turbulent). Thus, we expect the convection heat transfer relations to be rather complex because of the dependence of convection on so many variables. This is not surprising, since convection is the most complex mechanism of heat transfer.

Despite the complexity of convection, the rate of convection heat transfer is observed to be proportional to the temperature difference and is conveniently expressed by **Newton's law of cooling** as

$$\dot{q}_{\text{conv}} = h(T_s - T_\infty) \quad (\text{W/m}^2) \quad (6-1)$$

or

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty) \quad (\text{W}) \quad (6-2)$$

where

h = convection heat transfer coefficient, $\text{W}/\text{m}^2 \cdot \text{K}$

A_s = heat transfer surface area, m^2

T_s = temperature of the surface, $^\circ\text{C}$

T_∞ = temperature of the fluid sufficiently far from the surface, $^\circ\text{C}$

Judging from its units, the **convection heat transfer coefficient** h can be defined as *the rate of heat transfer between a solid surface and a fluid per unit surface area per unit temperature difference*.

You should not be deceived by the simple appearance of this relation, because the convection heat transfer coefficient h depends on several of the mentioned variables, and thus is difficult to determine.

Fluid flow is often confined by solid surfaces, and it is important to understand how the presence of solid surfaces affects fluid flow. Consider the flow of a fluid in a stationary pipe or over a solid surface that is nonporous (i.e., impermeable to the fluid). All experimental observations indicate that a fluid in motion comes to a complete stop at the surface and assumes a zero velocity relative to the surface. That is, a fluid in direct contact with a solid “sticks” to the surface due to viscous effects, and there is no slip. This is known as the **no-slip condition**.

Figure 6–3, obtained from a video clip, clearly shows the evolution of a velocity gradient as a result of the fluid sticking to the surface of a blunt nose. The layer that sticks to the surface slows the adjacent fluid layer because of viscous forces between the fluid layers, which slows the next layer, and so on. Therefore, the no-slip condition is responsible for the development of the velocity profile. The flow region adjacent to the wall in which the viscous effects (and thus the velocity gradients) are significant is called the **boundary layer**. The fluid property responsible for the no-slip condition and the development of the boundary layer is *viscosity* and is discussed briefly in Sec. 6–2.

A fluid layer adjacent to a moving surface has the same velocity as the surface. A consequence of the no-slip condition is that all velocity profiles must have zero values with respect to the surface at the points of contact between a fluid and a solid surface (Fig. 6–4). Another consequence of the no-slip condition is the *surface drag*, which is the force a fluid exerts on a surface in the flow direction.

An implication of the no-slip condition is that heat transfer from the solid surface to the fluid layer adjacent to the surface is by *pure conduction*, since the fluid layer is motionless, and it can be expressed as

$$\dot{q}_{\text{conv}} = \dot{q}_{\text{cond}} = -k_{\text{fluid}} \left. \frac{\partial T}{\partial y} \right|_{y=0} \quad (\text{W}/\text{m}^2) \quad (6-3)$$

where T represents the temperature distribution in the fluid and $(\partial T/\partial y)_{y=0}$ is the *temperature gradient* at the surface. Heat is then *convected* away from the surface as a result of fluid motion. Note that convection heat transfer from a solid surface to a fluid is merely the conduction heat transfer from the solid surface to the fluid layer adjacent to the surface. Therefore, we can equate Eqs. 6–1 and 6–3 for the heat flux to obtain

$$h = \frac{-k_{\text{fluid}}(\partial T/\partial y)_{y=0}}{T_s - T_\infty} \quad (\text{W}/\text{m}^2 \cdot \text{K}) \quad (6-4)$$

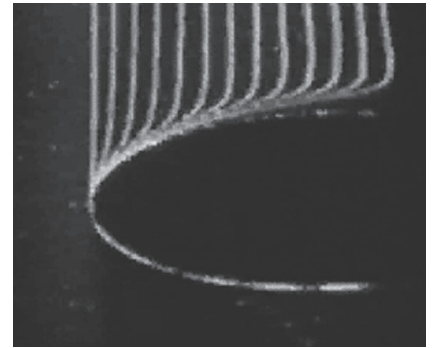


FIGURE 6–3

The development of a velocity profile due to the no-slip condition as a fluid flows over a blunt nose.

Courtesy of IIHR-Hydroscience & Engineering

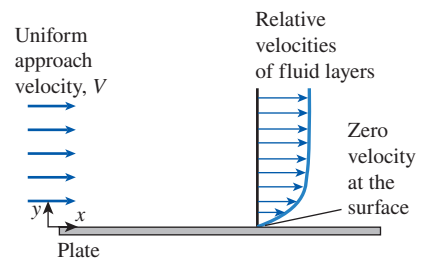


FIGURE 6–4

A fluid flowing over a stationary surface comes to a complete stop at the surface because of the no-slip condition.

**FIGURE 6-5**

Wilhelm Nusselt (1882–1957), was a German engineer, born in Nuremberg, Germany. He studied machinery at the Technical Universities of Berlin-Charlottenburg and Munchen and conducted advanced studies in mathematics and physics. His doctoral thesis was on the “Conductivity of Insulating Materials,” which he completed in 1907. In 1915, Nusselt published his pioneering paper: “The Basic Laws of Heat Transfer,” in which he first proposed the dimensionless groups now known as the principal parameters in the similarity theory of heat transfer. His other famous works were concerned with the film condensation of steam on vertical surfaces, the combustion of pulverized coal, and the analogy between heat and mass transfer in evaporation. Among his well-known mathematical works are the solutions for laminar heat transfer in the entrance region of tubes and for heat exchange in crossflow, and the basic theory of regenerators.

Source: Wikipedia

for the determination of the *convection heat transfer coefficient* when the temperature distribution within the fluid is known.

The convection heat transfer coefficient, in general, varies along the flow (or x -) direction. The *average* or *mean* convection heat transfer coefficient for a surface in such cases is determined by properly averaging the *local* convection heat transfer coefficients over the entire surface.

Nusselt Number

In convection studies, it is common practice to nondimensionalize the governing equations and combine the variables, which group together into *dimensionless numbers* in order to reduce the number of total variables. It is also common practice to nondimensionalize the heat transfer coefficient h with the Nusselt number, defined as

$$\text{Nu} = \frac{hL_c}{k} \quad (6-5)$$

where k is the thermal conductivity of the fluid and L_c is the *characteristic length*. The Nusselt number is named after Wilhelm Nusselt (Fig. 6–5), who made significant contributions to convective heat transfer in the first half of the 20th century, and it is viewed as the *dimensionless convection heat transfer coefficient*.

To understand the physical significance of the Nusselt number, consider a fluid layer of thickness L and temperature difference $\Delta T = T_2 - T_1$, as shown in Fig. 6–6. Heat transfer through the fluid layer is by *convection* when the fluid involves some motion and by *conduction* when the fluid layer is motionless. Heat flux (the rate of heat transfer per unit surface area) in either case is

$$\dot{q}_{\text{conv}} = h\Delta T \quad (6-6)$$

and

$$\dot{q}_{\text{cond}} = k \frac{\Delta T}{L} \quad (6-7)$$

Taking their ratio gives

$$\frac{\dot{q}_{\text{conv}}}{\dot{q}_{\text{cond}}} = \frac{h\Delta T}{k\Delta T/L} = \frac{hL}{k} = \text{Nu} \quad (6-8)$$

which is the Nusselt number. Therefore, the Nusselt number represents the enhancement of heat transfer through a fluid layer as a result of convection relative to conduction across the same fluid layer. The larger the Nusselt number, the more effective the convection. A Nusselt number of $\text{Nu} = 1$ for a fluid layer represents heat transfer across the layer by pure conduction.

We use forced convection in daily life more often than you might think (Fig. 6–7). We resort to forced convection whenever we want to increase the rate of heat transfer from a hot object. For example, we turn on the fan on hot summer days to help our bodies to cool more effectively. The higher the fan speed, the better we feel. We *stir* our soup and *blow* on a hot slice of pizza to make them cool faster. The air on windy winter days feels much colder than it actually is. The simplest solution to heating problems in electronics packaging is to use a large enough fan.

EXAMPLE 6–1 Heat Transfer Calculation from Temperature Profile

During the flow of air at $T_\infty = 20^\circ\text{C}$ over a plate surface maintained at a constant temperature of $T_s = 160^\circ\text{C}$, the dimensionless temperature profile within the air layer over the plate is determined to be

$$\frac{T(y) - T_\infty}{T_s - T_\infty} = e^{-ay}$$

where $a = 3200 \text{ m}^{-1}$ and y is the vertical distance measured from the plate surface in m (Fig. 6–8). Determine the heat flux on the plate surface and the convection heat transfer coefficient.

SOLUTION Airflow over a flat plate has a given temperature profile. The heat flux on the plate surface and the convection heat transfer coefficient are to be determined.

Assumptions 1 The given nondimensional temperature profile is representative of the variation of temperature over the entire plate. 2 Heat transfer by radiation is negligible.

Properties The thermal conductivity of air at the film temperature of $T_f = (T_s + T_\infty)/2 = (160^\circ\text{C} + 20^\circ\text{C})/2 = 90^\circ\text{C}$ is $k = 0.03024 \text{ W/m}\cdot\text{K}$ (Table A–15).

Analysis Noting that heat transfer from the plate to air at the surface is by conduction, heat flux from the solid surface to the fluid layer adjacent to the surface is determined from

$$\dot{q} = \dot{q}_{\text{cond}} = -k_{\text{fluid}} \left. \frac{\partial T}{\partial y} \right|_{y=0}$$

where the temperature gradient at the plate surface is

$$\begin{aligned} \left. \frac{\partial T}{\partial y} \right|_{y=0} &= (T_s - T_\infty)a[e^{-ay}]_{y=0} = (T_s - T_\infty)(-a) \\ &= (160^\circ\text{C} - 20^\circ\text{C})(-3200 \text{ m}^{-1}) = -4.48 \times 10^5 \text{ }^\circ\text{C/m} \end{aligned}$$

Substituting, the heat flux is determined to be

$$\dot{q} = -(0.03024 \text{ W/m}\cdot\text{K})(-4.48 \times 10^5 \text{ }^\circ\text{C/m}) = \mathbf{1.35 \times 10^4 \text{ W/m}^2}$$

Then the convection heat transfer coefficient becomes

$$\begin{aligned} h &= \frac{-k_{\text{fluid}}(\partial T/\partial y)_{y=0}}{T_s - T_\infty} \\ &= \frac{-(0.03024 \text{ W/m}\cdot\text{K})(-4.48 \times 10^5 \text{ }^\circ\text{C/m})}{(160^\circ\text{C} - 20^\circ\text{C})} = \mathbf{96.8 \text{ W/m}^2\cdot\text{K}} \end{aligned}$$

Discussion The convection heat transfer coefficient could also be determined from Newton's law of cooling, $\dot{q} = h(T_s - T_\infty)$.

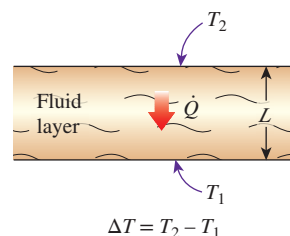


FIGURE 6–6

Heat transfer through a fluid layer of thickness L and temperature difference ΔT .

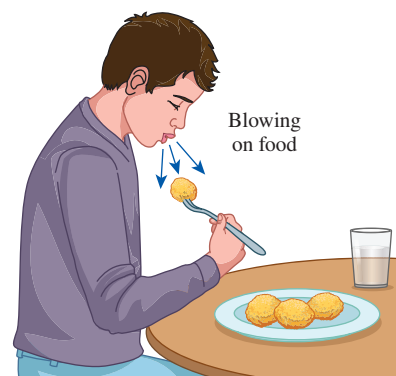


FIGURE 6–7

We resort to forced convection whenever we need to increase the rate of heat transfer.

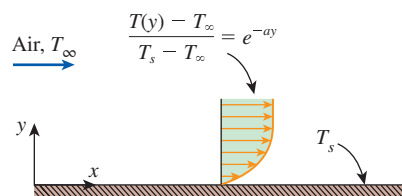


FIGURE 6–8

Schematic for Example 6–1.

6-2 ■ CLASSIFICATION OF FLUID FLOWS

Convection heat transfer is closely tied with fluid mechanics, which is the science that deals with the behavior of fluids at rest or in motion, and the interaction of fluids with solids or other fluids at the boundaries. A wide variety of fluid flow problems is encountered in practice, and it is usually convenient to classify them on the basis of some common characteristics to make it feasible to study them in groups. There are many ways to classify fluid flow problems, and here we present some general categories.

Viscous Versus Inviscid Regions of Flow

When two fluid layers move relative to each other, a friction force develops between them, and the slower layer tries to slow down the faster layer. This internal resistance to flow is quantified by the fluid property *viscosity*, which is a measure of internal stickiness of the fluid. Viscosity is caused by cohesive forces between the molecules in liquids and by molecular collisions in gases. There is no fluid with zero viscosity, and thus all fluid flows involve viscous effects to some degree. Flows in which the frictional effects are significant are called **viscous flows**. However, in many flows of practical interest, there are *regions* (typically regions not close to solid surfaces) where viscous forces are negligibly small compared to inertial or pressure forces. Neglecting the viscous terms in such **inviscid flow regions** greatly simplifies the analysis without much loss in accuracy.

The development of viscous and inviscid regions of flow as a result of inserting a flat plate parallel into a fluid stream of uniform velocity is shown in Fig. 6-9. The fluid sticks to the plate on both sides because of the no-slip condition, and the thin boundary layer in which the viscous effects are significant near the plate surface is the *viscous flow region*. The region of flow on both sides away from the plate and unaffected by the presence of the plate is the *inviscid flow region*.

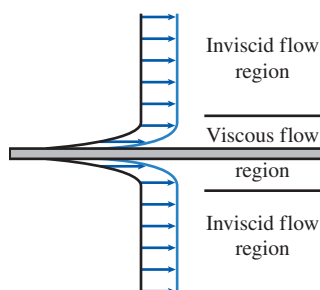


FIGURE 6-9

Flow of an originally uniform fluid stream over a flat plate, and the regions of viscous flow (next to the plate on both sides) and inviscid flow (away from the plate).

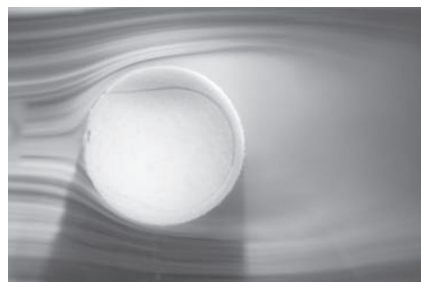


FIGURE 6-10

External flow over a tennis ball, and the turbulent wake region behind.

Courtesy NASA Ames Research Center and Cislunar Aerospace, Inc.

Internal Versus External Flow

A fluid flow is classified as being internal or external, depending on whether the fluid is forced to flow in a confined channel or over a surface. The flow of an unbounded fluid over a surface such as a plate, a wire, or a pipe is **external flow**. The flow in a pipe or duct is **internal flow** if the fluid is completely bounded by solid surfaces. Water flow in a pipe, for example, is internal flow, and airflow over a ball or over an exposed pipe during a windy day is external flow (Fig. 6-10). The flow of liquids in a duct is called *open-channel flow* if the duct is only partially filled with the liquid and there is a free surface. The flows of water in rivers and irrigation ditches are examples of such flows.

Internal flows are dominated by the influence of viscosity throughout the flow field. In external flows, the viscous effects are limited to boundary layers near solid surfaces and to wake regions downstream of bodies.

Compressible Versus Incompressible Flow

A flow is classified as being *compressible* or *incompressible*, depending on the level of variation of density during flow. Incompressibility is an approximation, and a flow is said to be **incompressible** if the density remains nearly

constant throughout. Therefore, the volume of every portion of fluid remains unchanged over the course of its motion when the flow (or the fluid) is incompressible.

The densities of liquids are essentially constant, and thus the flow of liquids is typically incompressible. Therefore, liquids are usually referred to as *incompressible substances*. A pressure of 210 atm, for example, causes the density of liquid water at 1 atm to change by just 1 percent. Gases, on the other hand, are highly compressible. A pressure change of just 0.01 atm, for example, causes a change of 1 percent in the density of atmospheric air.

Liquid flows are incompressible to a high level of accuracy, but the level of variation in density in gas flows and the consequent level of approximation made when modeling gas flows as incompressible depends on the Mach number, defined as $Ma = V/c$, where c is the **speed of sound** whose value is 346 m/s in air at room temperature at sea level. Gas flows can often be approximated as incompressible if the density changes are under about 5 percent, which is usually the case when $Ma < 0.3$. Therefore, the compressibility effects of air can be neglected at speeds under about 100 m/s. Note that the flow of a gas is not necessarily a compressible flow.

Small density changes of liquids corresponding to large pressure changes can still have important consequences. The irritating “water hammer” in a water pipe, for example, is caused by the vibrations of the pipe generated by the reflection of pressure waves following the sudden closing of the valves.

Laminar Versus Turbulent Flow

Some flows are smooth and orderly while others are rather chaotic. The highly ordered fluid motion characterized by smooth layers of fluid is called **laminar**. The word *laminar* comes from the movement of adjacent fluid particles together in “laminates.” The flow of high-viscosity fluids such as oils at low velocities is typically laminar. The highly disordered fluid motion that typically occurs at high velocities and is characterized by velocity fluctuations is called **turbulent** (Fig. 6–11). The flow of low-viscosity fluids such as air at high velocities is typically turbulent. The flow regime greatly influences the required power for pumping. A flow that alternates between being laminar and turbulent is called **transitional**.

Natural (or Unforced) Versus Forced Flow

A fluid flow is said to be natural or forced, depending on how the fluid motion is initiated. In **forced flow**, a fluid is forced to flow over a surface or in a pipe by external means such as a pump or a fan. In **natural flows**, any fluid motion is due to natural means such as the buoyancy effect, which manifests itself as the rise of the warmer (and thus lighter) fluid and the fall of cooler (and thus denser) fluid (Fig. 6–12). In solar hot-water systems, for example, the thermosiphoning effect is commonly used to replace pumps by placing the water tank sufficiently above the solar collectors.

Steady Versus Unsteady Flow

The terms *steady* and *uniform* are used frequently in engineering, and thus it is important to have a clear understanding of their meanings. The term **steady** implies *no change at a point with time*. The opposite of steady is **unsteady**.

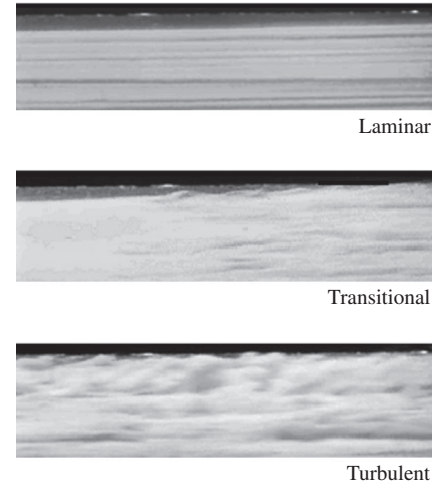


FIGURE 6–11

Laminar, transitional, and turbulent flows.

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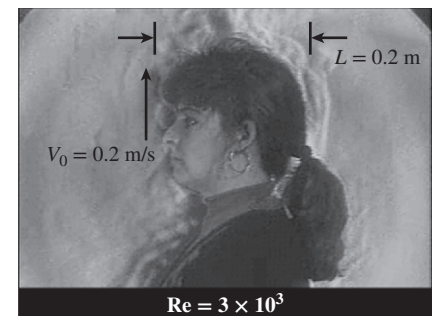


FIGURE 6–12

In this schlieren image, the rise of lighter, warmer air adjacent to the person's body indicates that humans and warm-blooded animals are surrounded by thermal plumes of rising warm air.

G.S. Settles, Gas Dynamics Lab, Penn State University. Used by permission.

The term **uniform** implies *no change with location* over a specified region. These meanings are consistent with their everyday use (steady girlfriend, uniform distribution, etc.).

The terms *unsteady* and *transient* are often used interchangeably, but these terms are not synonyms. In fluid mechanics, *unsteady* is the most general term that applies to any flow that is not steady, but **transient** is typically used for developing flows. When a rocket engine is fired up, for example, there are transient effects (the pressure builds up inside the rocket engine, the flow accelerates, etc.) until the engine settles down and operates steadily. The term **periodic** refers to the kind of unsteady flow in which the flow oscillates about a steady mean.

Many devices such as turbines, compressors, boilers, condensers, and heat exchangers operate for long periods of time under the same conditions, and they are classified as *steady-flow devices*. (Note that the flow field near the rotating blades of a turbomachine is of course unsteady, but we consider the overall flow field rather than the details at some localities when we classify devices.) During steady flow, the fluid properties can change from point to point within a device, but at any fixed point they remain constant. Therefore, the volume, the mass, and the total energy content of a steady-flow device or flow section remain constant in steady operation.

Steady-flow conditions can be closely approximated by devices that are intended for continuous operation such as turbines, pumps, boilers, condensers, and heat exchangers of power plants or refrigeration systems. Some cyclic devices, such as reciprocating engines or compressors, do not satisfy the steady-flow conditions since the flow at the inlets and the exits is pulsating and not steady. However, the fluid properties vary with time in a periodic manner, and the flow through these devices can still be analyzed as a steady-flow process by using time-averaged values for the properties.

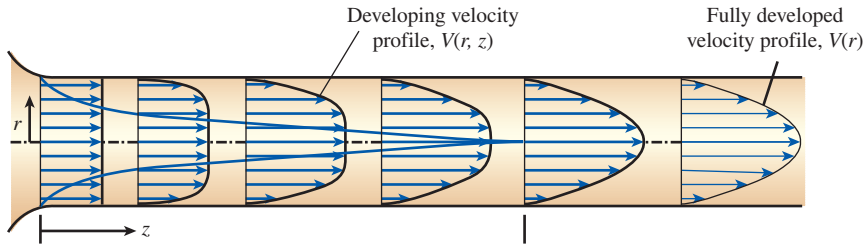
One-, Two-, and Three-Dimensional Flows

A flow field is best characterized by the velocity distribution, and thus a flow is said to be one-, two-, or three-dimensional if the flow velocity varies in one, two, or three primary dimensions, respectively. A typical fluid flow involves a three-dimensional geometry, and the velocity may vary in all three dimensions, rendering the flow three-dimensional [$\vec{V}(x, y, z)$ in rectangular or $\vec{V}(r, \theta, z)$ in cylindrical coordinates]. However, the variation of velocity in certain directions can be small relative to the variation in other directions and can be ignored with negligible error. In such cases, the flow can be modeled conveniently as being one- or two-dimensional, which is easier to analyze.

Consider steady flow of a fluid through a circular pipe attached to a large tank. The fluid velocity everywhere on the pipe surface is zero because of the no-slip condition, and the flow is two-dimensional in the entrance region of the pipe since the velocity changes in both the r - and z -directions. The velocity profile develops fully and remains unchanged after some distance from the inlet (about 10 pipe diameters in turbulent flow, and less in laminar pipe flow, as in Fig. 6–13), and the flow in this region is said to be *fully developed*. The fully developed flow in a circular pipe is *one-dimensional* since the velocity varies in the radial r -direction but not in the angular θ - or axial z -directions, as shown in Fig. 6–13. That is, the velocity profile is the same at any axial z -location, and it is symmetric about the axis of the pipe.

FIGURE 6–13

The development of the velocity profile in a circular pipe. $V = V(r, z)$ and thus the flow is two-dimensional in the entrance region, and it becomes one-dimensional downstream when the velocity profile fully develops and remains unchanged in the flow direction, $V = V(r)$.



Note that the dimensionality of the flow also depends on the choice of coordinate system and its orientation. The pipe flow discussed, for example, is one-dimensional in cylindrical coordinates, but two-dimensional in Cartesian coordinates—illustrating the importance of choosing the most appropriate coordinate system. Also note that even in this simple flow, the velocity cannot be uniform across the cross section of the pipe because of the no-slip condition. However, at a well-rounded entrance to the pipe, the velocity profile may be approximated as being nearly uniform across the pipe, since the velocity is nearly constant at all radii except very close to the pipe wall.

6–3 ■ VELOCITY BOUNDARY LAYER

Consider the parallel flow of a fluid over a *flat plate*, as shown in Fig. 6–14. Surfaces that are slightly contoured such as turbine blades can also be approximated as flat plates with reasonable accuracy. The x -coordinate is measured along the plate surface from the *leading edge* of the plate in the direction of the flow, and y is measured from the surface in the normal direction. The fluid approaches the plate in the x -direction with a uniform velocity V , which is practically identical to the free-stream velocity over the plate away from the surface (this would not be the case for crossflow over blunt bodies such as a cylinder).

For the sake of discussion, we can consider the fluid to consist of adjacent layers piled on top of each other. The velocity of the particles in the first fluid

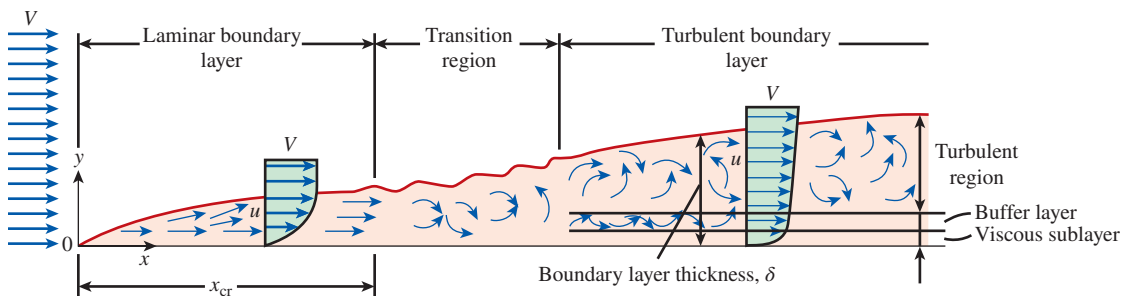
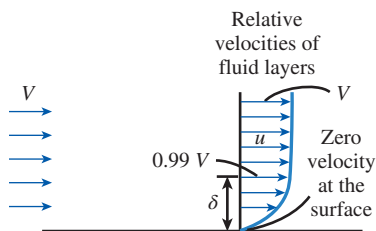


FIGURE 6–14

The development of the boundary layer for flow over a flat plate, and the different flow regimes.

Courtesy of University of Delaware.

**FIGURE 6-15**

The development of a boundary layer on a surface is due to the no-slip condition and friction.

layer adjacent to the plate becomes zero because of the no-slip condition. This motionless layer slows down the particles of the neighboring fluid layer as a result of friction between the particles of these two adjoining fluid layers at different velocities. This fluid layer then slows down the molecules of the next layer, and so on. Thus, the presence of the plate is felt up to some normal distance δ from the plate beyond which the free-stream velocity remains essentially unchanged. As a result, the x -component of the fluid velocity, u , varies from 0 at $y = 0$ to nearly V at $y = \delta$ (Fig. 6-15).

The region of the flow above the plate bounded by δ in which the effects of the viscous shearing forces caused by fluid viscosity are felt is called the **velocity boundary layer**. The *boundary layer thickness*, δ , is typically defined as the distance y from the surface at which $u = 0.99V$.

The hypothetical line of $u = 0.99V$ divides the flow over a plate into two regions: the **boundary layer region**, in which the viscous effects and the velocity changes are significant, and the **irrotational flow region**, in which the frictional effects are negligible and the velocity remains essentially constant.

Wall Shear Stress

Consider the flow of a fluid over the surface of a plate. The fluid layer in contact with the surface tries to drag the plate along via friction, exerting a *friction force* on it. Likewise, a faster fluid layer tries to drag the adjacent slower layer and exert a friction force because of the friction between the two layers. Friction force per unit area is called **shear stress** and is denoted by τ . Experimental studies indicate that the shear stress for most fluids is proportional to the *velocity gradient*, and the shear stress at the wall surface is expressed as

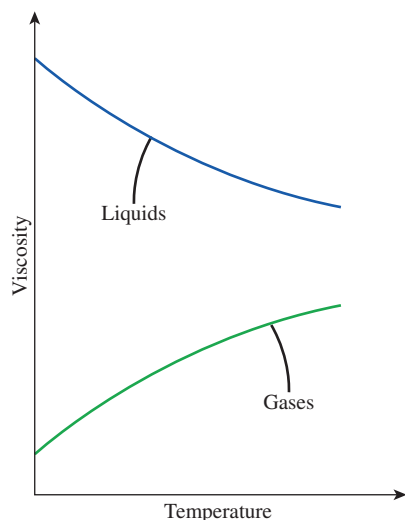
$$\tau_w = \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} \quad (\text{N/m}^2) \quad (6-9)$$

where the constant of proportionality μ is the **dynamic viscosity** of the fluid, whose unit is $\text{kg/m}\cdot\text{s}$ (or equivalently, $\text{N}\cdot\text{s/m}^2$, or $\text{Pa}\cdot\text{s}$, or poise = $0.1 \text{ Pa}\cdot\text{s}$).

The fluids that obey the linear relationship above are called **Newtonian fluids**, after Sir Isaac Newton, who expressed it first in 1687 (Fig. 1-36). Most common fluids such as water, air, gasoline, and oils are Newtonian fluids. Blood and liquid plastics are examples of non-Newtonian fluids. In this text we consider Newtonian fluids only.

In fluid flow and heat transfer studies, the ratio of dynamic viscosity to density appears frequently. For convenience, this ratio is given the name **kinematic viscosity** ν and is expressed as $\nu = \mu/\rho$. Two common units of kinematic viscosity are m^2/s and *stoke* ($1 \text{ stoke} = 1 \text{ cm}^2/\text{s} = 0.0001 \text{ m}^2/\text{s}$).

The viscosity of a fluid is a measure of its *resistance to deformation*, and it is a strong function of temperature. The viscosities of liquids *decrease* with temperature, whereas the viscosities of gases *increase* with temperature (Fig. 6-16). The viscosities of some fluids at 20°C are listed in Table 6-1. Note that the viscosities of different fluids differ by several orders of magnitude.

**FIGURE 6-16**

The viscosity of liquids decreases and the viscosity of gases increases with temperature.

The determination of the wall shear stress τ_w from Eq. 6–9 is not practical since it requires a knowledge of the flow velocity profile. A more practical approach in external flow is to relate τ_w to the upstream velocity V as

$$\tau_w = C_f \frac{\rho V^2}{2} \quad (\text{N/m}^2) \quad (6-10)$$

where C_f is the dimensionless **friction coefficient** or **skin friction coefficient**, whose value in most cases is determined experimentally, and ρ is the density of the fluid. Note that the friction coefficient, in general, varies with location along the surface. Once the average friction coefficient over a given surface is available, the friction force over the entire surface is determined from

$$F_f = C_f A_s \frac{\rho V^2}{2} \quad (\text{N}) \quad (6-11)$$

where A_s is the surface area.

The friction coefficient is an important parameter in heat transfer studies since it is directly related to the heat transfer coefficient and the power requirements of the pump or fan.

6-4 ■ THERMAL BOUNDARY LAYER

We have seen that a velocity boundary layer develops when a fluid flows over a surface as a result of the fluid layer adjacent to the surface assuming the surface velocity (i.e., zero velocity relative to the surface). Also, we defined the velocity boundary layer as the region in which the fluid velocity varies from zero to $0.99V$. Likewise, a *thermal boundary layer* develops when a fluid at a specified temperature flows over a surface that is at a different temperature, as shown in Fig. 6–17.

Consider the flow of a fluid at a uniform temperature of T_∞ over an isothermal flat plate at temperature T_s . The fluid particles in the layer adjacent to the surface reach thermal equilibrium with the plate and assume the surface temperature T_s . These fluid particles then exchange energy with the particles in the adjoining fluid layer, and so on. As a result, a temperature profile develops in the flow field that ranges from T_s at the surface to T_∞ sufficiently far from the surface. The flow region over the surface in which the temperature variation in the direction normal to the surface is significant is the **thermal boundary layer**. The *thickness* of the thermal boundary layer δ_t at any location along the surface is defined as *the distance from the surface at which the temperature difference $T - T_s$ equals $0.99(T_\infty - T_s)$* . Note that for the special case of $T_s = 0$, we have $T = 0.99T_\infty$ at the outer edge of the thermal boundary layer, which is analogous to $u = 0.99V$ for the velocity boundary layer.

The thickness of the thermal boundary layer increases in the flow direction, since the effects of heat transfer are felt at greater distances from the surface further downstream.

The convection heat transfer rate anywhere along the surface is directly related to the temperature gradient at that location. Therefore, the shape of the temperature profile in the thermal boundary layer dictates the convection heat transfer between a solid surface and the fluid flowing over it. In flow over a heated (or cooled) surface, both velocity and thermal boundary layers develop

TABLE 6–1

Dynamic viscosities of some fluids at 1 atm and 20°C (unless otherwise stated)

Fluid	Dynamic Viscosity μ , kg/m·s
Glycerin:	
–20°C	134.0
0°C	10.5
20°C	1.52
40°C	0.31
Engine oil:	
SAE 10W	0.10
SAE 10W30	0.17
SAE 30	0.29
SAE 50	0.86
Mercury	0.0015
Ethyl alcohol	0.0012
Water:	
0°C	0.0018
20°C	0.0010
100°C (liquid)	0.00028
100°C (vapor)	0.000012
Blood, 37°C	0.00040
Gasoline	0.00029
Ammonia	0.00015
Air	0.000018
Hydrogen, 0°C	0.0000088

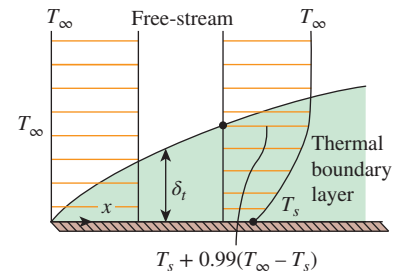


FIGURE 6–17

Thermal boundary layer on a flat plate (the fluid is hotter than the plate surface).

**FIGURE 6–18**

Ludwig Prandtl (1875–1953), was a German physicist famous for his work in aeronautics, born in Freising, Bavaria. His discovery in 1904 of the boundary layer which adjoins the surface of a body moving in a fluid led to an understanding of skin friction drag and of the way in which streamlining reduces the drag of airplane wings and other moving bodies. Prandtl's work and decisive advances in boundary layer and wing theories became the basic material of aeronautics. He also made important contributions to the theories of supersonic flow and of turbulence, and he contributed much to the development of wind tunnels and other aerodynamic equipment. The dimensionless **Prandtl number** was named after him.

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simultaneously. Noting that the fluid velocity has a strong influence on the temperature profile, the development of the velocity boundary layer relative to the thermal boundary layer will have a strong effect on the convection heat transfer.

Prandtl Number

The relative thickness of the velocity and the thermal boundary layers is best described by the *dimensionless* parameter **Prandtl number**, defined as

$$\text{Pr} = \frac{\text{Molecular diffusivity of momentum}}{\text{Molecular diffusivity of heat}} = \frac{\nu}{\alpha} = \frac{\mu c_p}{k} \quad (6-12)$$

It is named after Ludwig Prandtl (Fig. 6–18), who introduced the concept of boundary layer in 1904 and made significant contributions to boundary layer theory. The Prandtl numbers of fluids range from less than 0.01 for liquid metals to more than 100,000 for heavy oils (Table 6–2). Note that the Prandtl number is in the order of 10 for water.

The Prandtl numbers of gases are about 1, which indicates that both momentum and heat dissipate through the fluid at about the same rate. Heat diffuses very quickly in liquid metals ($\text{Pr} \ll 1$) and very slowly in oils ($\text{Pr} \gg 1$) relative to momentum. Consequently the thermal boundary layer is much thicker for liquid metals and much thinner for oils relative to the velocity boundary layer.

Liquid metals are a special class of fluids with very low Prandtl numbers (Table 6–2). The very low Prandtl number is due to the high thermal conductivity of these fluids, since the specific heat and viscosity of liquid metals are very comparable to other common fluids. Considerable interest has been placed on liquid metals as coolants in applications where large amounts of heat must be removed from a relatively small space, as in a nuclear reactor. Liquid metals, aside from having high thermal conductivity values, have high thermal capacity, low vapor pressure, and low melting point. They remain in the liquid state at higher temperatures than conventional fluids. This makes them more attractive for use in compact heat exchangers. However, liquid metals are corrosive in nature, and their contact with air or water may result in violent action. Suitable measures for handling them have been developed.

6–5 ■ LAMINAR AND TURBULENT FLOWS

If you have been around smokers, you probably noticed that the cigarette smoke rises in a smooth plume for the first few centimeters and then starts fluctuating randomly in all directions as it continues its rise. Other plumes behave similarly (Fig. 6–19). Likewise, a careful inspection of flow in a pipe reveals that the fluid flow is streamlined at low velocities but turns chaotic as the velocity is increased above a critical value, as shown in Fig. 6–20. The flow regime in the first case is said to be **laminar**, characterized by *smooth streamlines* and *highly ordered motion*, and **turbulent** in the second case, where it is characterized by *velocity fluctuations* and *highly disordered motion*. The **transition** from laminar to turbulent flow does not occur suddenly; rather, it occurs over some region in which the flow fluctuates between laminar and turbulent flows before it becomes fully turbulent. Most flows encountered in practice are turbulent. Laminar flow is encountered when highly viscous fluids such as oils flow in small pipes or narrow passages.

TABLE 6–2

Typical ranges of Prandtl numbers for common fluids

Fluid	Pr
Liquid metals	0.004–0.030
Gases	0.7–1.0
Water	1.7–13.7
Light organic fluids	5–50
Oils	50–100,000
Glycerin	2000–100,000

We can verify the existence of these laminar, transitional, and turbulent flow regimes by injecting some dye streak into the flow in a glass tube, as the British scientist Osborn Reynolds (Fig. 6–21) did over a century ago. We observe that the dye streak forms a *straight and smooth line* at low velocities when the flow is laminar (we may see some blurring because of molecular diffusion), has *bursts of fluctuations* in the transitional regime, and *zigzags rapidly and randomly* when the flow becomes fully turbulent. These zigzags and the dispersion of the dye are indicative of the fluctuations in the main flow and the rapid mixing of fluid particles from adjacent layers.

Typical average velocity profiles in laminar and turbulent flow are also given in Fig. 6–14. Note that the velocity profile in turbulent flow is much fuller than that in laminar flow, with a sharp drop near the surface. The turbulent boundary layer can be considered to consist of four regions, characterized by the distance from the wall. The very thin layer next to the wall where viscous effects are dominant is the **viscous sublayer**. The velocity profile in this layer is very nearly *linear*, and the flow is streamlined. Next to the viscous sublayer is the **buffer layer**, in which turbulent effects are becoming significant, but the flow is still dominated by viscous effects. Above the buffer layer is the **overlap layer**, in which the turbulent effects are much more significant, but still not dominant. Above that is the **turbulent layer**, in which turbulent effects dominate over viscous effects.

The *intense mixing* of the fluid in turbulent flow as a result of rapid fluctuations enhances heat and momentum transfer between fluid particles, which increases the friction force on the surface and the convection heat transfer rate. It also causes the boundary layer to enlarge. Both the friction and heat transfer coefficients reach maximum values when the flow becomes *fully turbulent*. So it will come as no surprise that a special effort is made in the design of heat transfer coefficients associated with turbulent flow. The enhancement in heat transfer in turbulent flow does not come for free, however. It may be necessary to use a larger pump to overcome the larger friction forces accompanying the higher heat transfer rate.

Reynolds Number

The transition from laminar to turbulent flow depends on the *surface geometry, surface roughness, flow velocity, surface temperature, and type of fluid*, among other things. After exhaustive experiments in the 1880s, Osborne Reynolds discovered that the flow regime depends mainly on the ratio of the *inertia forces* to *viscous forces* in the fluid. This ratio is called the **Reynolds number** (Fig. 6–21), which is a *dimensionless* quantity, and which is expressed for external flow as (Fig. 6–22)

$$\text{Re} = \frac{\text{Inertia forces}}{\text{Viscous forces}} = \frac{VL_c}{\nu} = \frac{\rho VL_c}{\mu} \quad (6-13)$$

where V is the upstream velocity (equivalent to the free-stream velocity for a flat plate), L_c is the characteristic length of the geometry, and $\nu = \mu/\rho$ is the kinematic viscosity of the fluid. For a flat plate, the characteristic length is the distance x from the leading edge. Note that kinematic viscosity has the unit m^2/s , which is identical to the unit of thermal diffusivity and which can be viewed as *viscous diffusivity* or *diffusivity for momentum*.

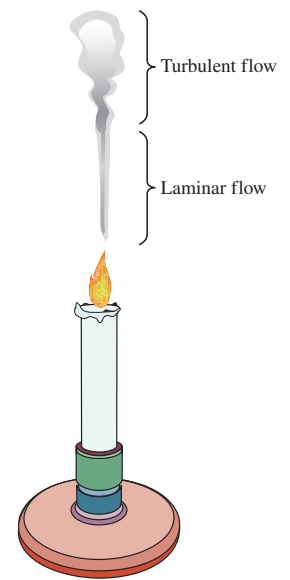


FIGURE 6–19
Laminar and turbulent flow regimes of candle smoke.

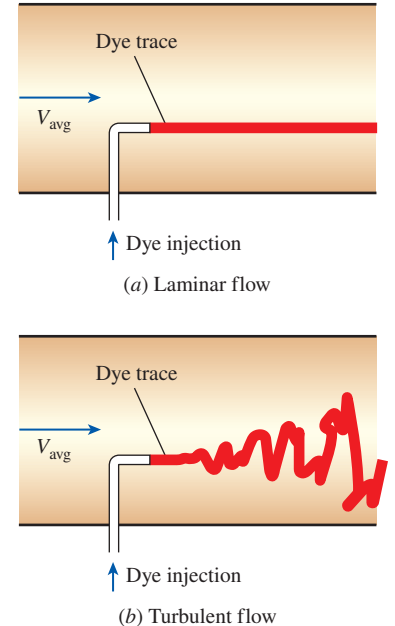


FIGURE 6–20
The behavior of colored fluid injected into the flow in laminar and turbulent flows in a pipe.

**FIGURE 6–21**

Osborne Reynolds (1842–1912), an English engineer and physicist best known for his work in the fields of hydraulics and hydrodynamics, was born in Belfast, Ireland. Reynolds' studies of condensation and the transfer of heat between solids and fluids brought about radical revisions in boiler and condenser design, and his work on turbine pumps laid the foundation for their rapid development. His classical paper on "The Law of Resistance in Parallel Channels" (1883) investigated the transition from smooth, or laminar, to turbulent flow. In 1886 he also formulated "The Theory of Lubrication," and later in 1889, he developed a mathematical framework which became the standard in turbulence work. His other work included the explanation of the radiometer and an early absolute determination of the mechanical equivalent of heat. The dimensionless **Reynolds number**, which provides a criterion for dynamic similarity and for correct modeling in many fluid flow experiments, is named after him.

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At *large* Reynolds numbers, the inertia forces, which are proportional to the density and the velocity of the fluid, are large relative to the viscous forces, and thus the viscous forces cannot prevent the random and rapid fluctuations of the fluid. At *small* or *moderate* Reynolds numbers, however, the viscous forces are large enough to suppress these fluctuations and to keep the fluid "in line." Thus the flow is *turbulent* in the first case and *laminar* in the second.

The Reynolds number at which the flow becomes turbulent is called the **critical Reynolds number**. The value of the critical Reynolds number is different for different geometries and flow conditions. For flow over a flat plate, the generally accepted value of the critical Reynolds number is $Re_{cr} = Vx_{cr}/\nu = 5 \times 10^5$, where x_{cr} is the distance from the leading edge of the plate at which transition from laminar to turbulent flow occurs. The value of Re_{cr} may change substantially, however, depending on the level of turbulence in the free stream.

6–6 ■ HEAT AND MOMENTUM TRANSFER IN TURBULENT FLOW

Most flows encountered in engineering practice are turbulent, and thus it is important to understand how turbulence affects wall shear stress and heat transfer. However, turbulent flow is a complex mechanism dominated by fluctuations, and despite tremendous amounts of work done in this area by researchers, the theory of turbulent flow is still not fully understood. Therefore, we must rely on experiments and the empirical or semi-empirical correlations developed for various situations.

Turbulent flow is characterized by disorderly and rapid fluctuations of swirling regions of fluid, called **eddies**, throughout the flow. These fluctuations provide an additional mechanism for momentum and energy transfer. In laminar flow, fluid particles flow in an orderly manner along pathlines, and momentum and energy are transferred across streamlines by molecular diffusion. In turbulent flow, the swirling eddies transport mass, momentum, and energy to other regions of flow much more rapidly than molecular diffusion, greatly enhancing mass, momentum, and heat transfer. As a result, turbulent flow is associated with much higher values of friction, heat transfer, and mass transfer coefficients (Fig. 6–23).

Even when the average flow is steady, the eddy motion in turbulent flow causes significant fluctuations in the values of velocity, temperature, pressure, and even density (in compressible flow). Figure 6–24 shows the variation of the instantaneous velocity component u with time at a specified location, as can be measured with a hot-wire anemometer probe or other sensitive device. We observe that the instantaneous values of the velocity fluctuate about an average value, which suggests that the velocity can be expressed as the sum of an *average value* $\bar{u}$ and a *fluctuating component* u' ,

$$u = \bar{u} + u' \quad (6-14)$$

This is also the case for other properties such as the velocity component v in the y -direction, and thus $v = \bar{v} + v'$, $P = \bar{P} + P'$, and $T = \bar{T} + T'$. The average value of a property at some location is determined by averaging it over a time interval that is sufficiently large so that the time average levels off to a constant. Therefore, the time average of fluctuating components is zero, e.g., $\bar{u'} = 0$. The magnitude of u' is usually just a few percent of $\bar{u}$, but the high frequencies of eddies (in the order of a thousand per second) makes

them very effective for the transport of momentum, thermal energy, and mass. In time-averaged *stationary* turbulent flow, the average values of properties (indicated by an overbar) are independent of time. The chaotic fluctuations of fluid particles play a dominant role in pressure drop, and these random motions must be considered in analyses together with the average velocity.

Perhaps the first thought that comes to mind is to determine the shear stress in an analogous manner to laminar flow from $\tau = -\mu \, d\bar{u}/dr$, where $\bar{u}(r)$ is the average velocity profile for turbulent flow. But the experimental studies show that this is not the case, and the shear stress is much larger due to the turbulent fluctuations. Therefore, it is convenient to think of the turbulent shear stress as consisting of two parts: the *laminar component*, which accounts for the friction between layers in the flow direction (expressed as $\tau_{\text{lam}} = \mu(d\bar{u}/dr)$, and the *turbulent component*, which accounts for the friction between the fluctuating fluid particles and the fluid body (denoted as τ_{turb} and is related to the fluctuation components of velocity).

Consider turbulent flow in a horizontal pipe and the upward eddy motion of fluid particles in a layer of lower velocity to an adjacent layer of higher velocity through a differential area dA as a result of the velocity fluctuation v' , as shown in Fig. 6–25. The mass flow rate of the fluid particles rising through dA is $\rho v' dA$, and its net effect on the layer above dA is a reduction in its average flow velocity because of momentum transfer to the fluid particles with lower average flow velocity. This momentum transfer causes the horizontal velocity of the fluid particles to increase by u' , and thus causes its momentum in the horizontal direction to increase at a rate of $(\rho v' dA)u'$, which must be equal to the decrease in the momentum of the upper fluid layer.

Noting that force in a given direction is equal to the rate of change of momentum in that direction, the horizontal force acting on a fluid element above dA due to the passing of fluid particles through dA is $\delta F = (\rho v' dA)(-u') = -\rho u' v' dA$. Therefore, the shear force per unit area due to the eddy motion of fluid particles $\delta F/dA = -\rho u' v'$ can be viewed as the instantaneous turbulent shear stress. Then the **turbulent shear stress** can be expressed as $\tau_{\text{turb}} = -\rho \overline{u' v'}$ where $\overline{u' v'}$ is the time average of the product of the fluctuating velocity components u' and v' . Similarly, considering that $h = c_p T$ represents the energy of the fluid and T' is the eddy temperature relative to the mean value, the rate of thermal energy transport by turbulent eddies is $\dot{q}_{\text{turb}} = \rho c_p \overline{v' T'}$. Note that $\overline{u' v'} \neq 0$, even though $\overline{u'} = 0$ and $\overline{v'} = 0$ (and thus $\overline{u' v'} = 0$), and experimental results show that $\overline{u' v'}$ is usually a negative quantity. Terms such as $-\rho \overline{u' v'}$ or $-\rho \overline{u'^2}$ are called **Reynolds stresses** or **turbulent stresses**.

The random eddy motion of groups of particles resembles the random motion of molecules in a gas—colliding with each other after traveling a certain distance and exchanging momentum and heat in the process. Therefore, momentum and heat transport by eddies in turbulent boundary layers is analogous to the molecular momentum and heat diffusion. Then turbulent wall shear stress and turbulent heat transfer can be expressed in an analogous manner as

$$\tau_{\text{turb}} = -\rho \overline{u' v'} = \mu_t \frac{\partial \bar{u}}{\partial y} \quad \text{and} \quad \dot{q}_{\text{turb}} = \rho c_p \overline{v' T'} = -k_t \frac{\partial T}{\partial y} \quad (6-15)$$

where μ_t is called the **turbulent** (or **eddy**) **viscosity**, which accounts for momentum transport by turbulent eddies, and k_t is called the **turbulent** (or **eddy**) **thermal conductivity**, which accounts for thermal energy transport by

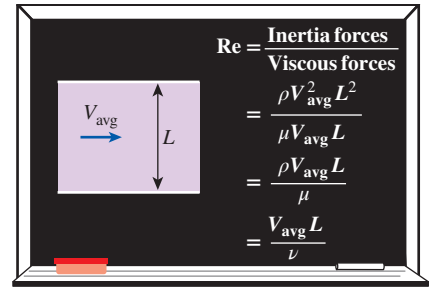


FIGURE 6–22

The Reynolds number can be viewed as the ratio of inertial forces to viscous forces acting on a fluid element.

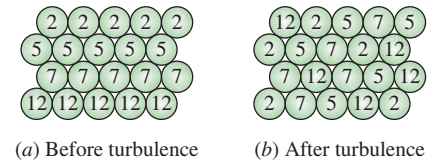


FIGURE 6–23

The intense mixing in turbulent flow brings fluid particles at different temperatures into close contact and thus enhances heat transfer.

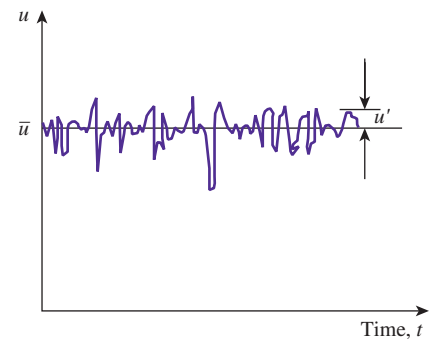


FIGURE 6–24

Fluctuations of the velocity component u with time at a specified location in turbulent flow.

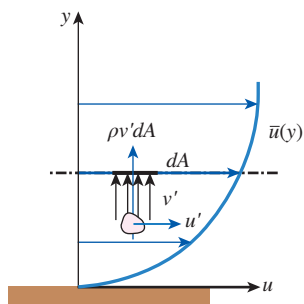


FIGURE 6-25

Fluid particle moving upward through a differential area dA as a result of the velocity fluctuation v' .

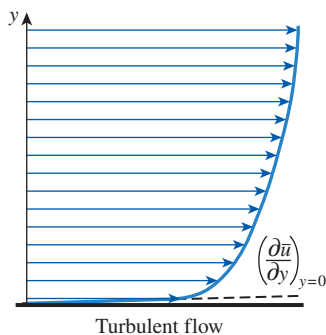
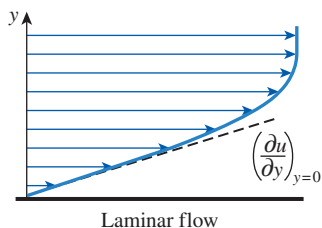


FIGURE 6-26

The velocity gradients at the wall, and thus the wall shear stress, are much larger for turbulent flow than they are for laminar flow, even though the turbulent boundary layer is thicker than the laminar one for the same value of free-stream velocity.

turbulent eddies. Then the total shear stress and total heat flux can be expressed conveniently as

$$\tau_{\text{total}} = (\mu + \mu_t) \frac{\partial \bar{u}}{\partial y} = \rho(\nu + \nu_t) \frac{\partial \bar{u}}{\partial y} \quad (6-16)$$

and

$$\dot{q}_{\text{total}} = -(k + k_t) \frac{\partial \bar{T}}{\partial y} = -\rho c_p (\alpha + \alpha_t) \frac{\partial \bar{T}}{\partial y} \quad (6-17)$$

where $\nu_t = \mu_t / \rho$ is the **kinematic eddy viscosity** (or **eddy diffusivity of momentum**) and $\alpha_t = k_t / \rho c_p$ is the **eddy thermal diffusivity** (or **eddy diffusivity of heat**).

Eddy motion and thus eddy diffusivities are much larger than their molecular counterparts in the core region of a turbulent boundary layer. The eddy motion loses its intensity close to the wall and diminishes at the wall because of the no-slip condition. Therefore, the velocity and temperature profiles are very slowly changing in the core region of a turbulent boundary layer, but they are very steep in the thin layer adjacent to the wall, resulting in large velocity and temperature gradients at the wall surface. So it is no surprise that the wall shear stress and wall heat flux are much larger in turbulent flow than they are in laminar flow (Fig. 6-26).

Note that molecular diffusivities ν and α (as well as μ and k) are fluid properties, and their values can be found listed in fluid handbooks. Eddy diffusivities ν_t and α_t (as well as μ_t and k_t), however, are *not* fluid properties, and their values depend on flow conditions. Eddy diffusivities ν_t and α_t decrease toward the wall, becoming zero at the wall. Their values range from zero at the wall to several thousand times the values of molecular diffusivities in the core region.

6-7 ■ DERIVATION OF DIFFERENTIAL CONVECTION EQUATIONS*

In this section we derive the governing equations of fluid flow in the boundary layers. To keep the analysis at a manageable level, we assume the flow to be steady and two-dimensional and the fluid to be Newtonian with constant properties (density, viscosity, thermal conductivity, etc.).

Consider the parallel flow of a fluid over a surface. We take the flow direction along the surface to be x and the direction normal to the surface to be y , and we choose a differential volume element of length dx , height dy , and unit depth in the z -direction (normal to the paper) for analysis (Fig. 6-27). The fluid flows over the surface with a uniform free-stream velocity V , but the velocity within the boundary layer is two-dimensional: the x -component of the velocity is u , and the y -component is v . Note that $u = u(x, y)$ and $v = v(x, y)$ in steady two-dimensional flow.

Next we apply three fundamental laws to this fluid element: Conservation of mass, conservation of momentum, and conservation of energy to obtain the continuity, momentum, and energy equations for laminar flow in boundary layers.

*This and the upcoming sections of this chapter deal with theoretical aspects of convection, and can be skipped and be used as a reference if desired without a loss in continuity.

The Continuity Equation

The conservation of mass principle is simply a statement that mass cannot be created or destroyed during a process and all the mass must be accounted for during an analysis. In steady flow, the amount of mass within the control volume remains constant, and thus the conservation of mass can be expressed as

$$\left(\begin{array}{c} \text{Rate of mass flow} \\ \text{into the control volume} \end{array} \right) = \left(\begin{array}{c} \text{Rate of mass flow} \\ \text{out of the control volume} \end{array} \right) \quad (6-18)$$

Noting that mass flow rate is equal to the product of density, average velocity, and cross-sectional area normal to flow, the rate at which fluid enters the control volume from the left surface is $\rho u(dy \cdot 1)$. The rate at which the fluid leaves the control volume from the right surface can be expressed as

$$\rho \left(u + \frac{\partial u}{\partial x} dx \right) (dy \cdot 1) \quad (6-19)$$

Repeating this for the y direction and substituting the results into Eq. 6-18, we obtain

$$\rho u(dy \cdot 1) + \rho v(dx \cdot 1) = \rho \left(u + \frac{\partial u}{\partial x} dx \right) (dy \cdot 1) + \rho \left(v + \frac{\partial v}{\partial y} dy \right) (dx \cdot 1) \quad (6-20)$$

Simplifying and dividing by $dx \cdot dy \cdot 1$ gives

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (6-21)$$

This is the *conservation of mass* relation in differential form, which is also known as the **continuity equation** or **mass balance** for steady two-dimensional flow of a fluid with constant density.

The Momentum Equations

The differential forms of the equations of motion in the velocity boundary layer are obtained by applying Newton's second law of motion to a differential control volume element in the boundary layer. Newton's second law is an expression for momentum balance and can be stated as *the net force acting on the control volume is equal to the mass times the acceleration of the fluid element within the control volume, which is also equal to the net rate of momentum outflow from the control volume*.

The forces acting on the control volume consist of *body forces* that act throughout the entire body of the control volume (such as gravity, electric, and magnetic forces) and are proportional to the volume of the body, and *surface forces* that act on the control surface (such as the pressure forces due to hydrostatic pressure and shear stresses due to viscous effects) and are proportional to the surface area. The surface forces appear as the control volume is isolated from its surroundings for analysis, and the effect of the detached body is replaced by a force at that location. Note that pressure represents the compressive force applied on the fluid element by the surrounding fluid, and it is always directed to the surface.

We express Newton's second law of motion for the control volume as

$$(\text{Mass}) \left(\begin{array}{c} \text{Acceleration} \\ \text{in a specified direction} \end{array} \right) = \left(\begin{array}{c} \text{Net force (body and surface)} \\ \text{acting in that direction} \end{array} \right) \quad (6-22)$$

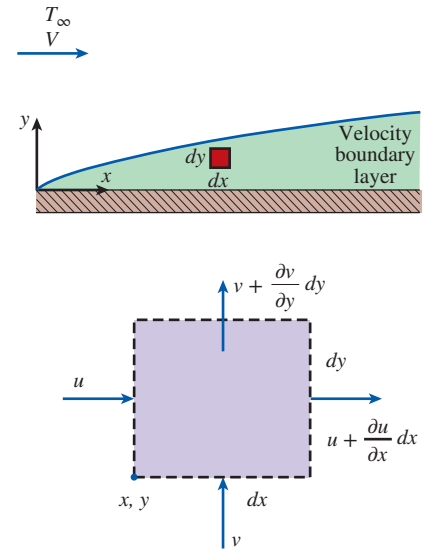


FIGURE 6-27

Differential control volume used in the derivation of mass balance in velocity boundary layer in two-dimensional flow over a surface.

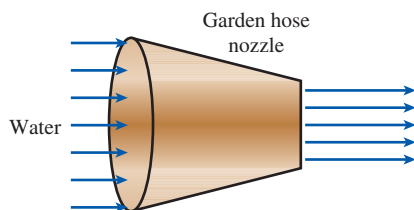


FIGURE 6-28

During steady flow, a fluid may not accelerate in time at a fixed point, but it may accelerate in space.

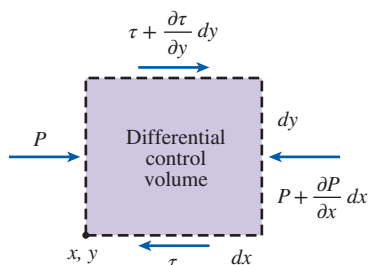


FIGURE 6-29

Differential control volume used in the derivation of x -momentum equation in velocity boundary layer in two-dimensional flow over a surface.

or

$$\delta m \cdot a_x = F_{\text{surface}, x} + F_{\text{body}, x} \quad (6-23)$$

where the mass of the fluid element within the control volume is

$$\delta m = \rho(dx \cdot dy \cdot 1) \quad (6-24)$$

Noting that flow is steady and two-dimensional and thus $u = u(x, y)$, the total differential of u is

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \quad (6-25)$$

Then the acceleration of the fluid element in the x -direction becomes

$$a_x = \frac{du}{dt} = \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \quad (6-26)$$

You may be tempted to think that acceleration is zero in steady flow since acceleration is the rate of change of velocity with time, and in steady flow there is no change with time. Well, a garden hose nozzle tells us that this understanding is not correct. Even in steady flow and thus constant mass flow rate, water accelerates through the nozzle (Fig. 6-28). *Steady* simply means no change with time at a specified location (and thus $\partial u / \partial t = 0$), but the value of a quantity may change from one location to another (and thus $\partial u / \partial x$ and $\partial u / \partial y$ may be different from zero). In the case of a nozzle, the velocity of water remains constant at a specified point, but it changes from the inlet to the exit (water accelerates along the nozzle, which is the reason for attaching a nozzle to the garden hose in the first place).

The forces acting on a surface are due to pressure and viscous effects. In two-dimensional flow, the *viscous stress* at any point on an imaginary surface within the fluid can be resolved into two perpendicular components: one normal to the surface called *normal stress* (which should not be confused with pressure) and another along the wall surface called *shear stress*. The normal stress is related to the velocity gradients $\partial u / \partial x$ and $\partial v / \partial y$, which are much smaller than $\partial u / \partial y$, to which shear stress is related. Neglecting the normal stresses for simplicity, the surface forces acting on the control volume in the x -direction are as shown in Fig. 6-29. Then the net surface force acting in the x -direction becomes

$$\begin{aligned} F_{\text{surface}, x} &= \left(\frac{\partial \tau}{\partial y} dy \right) (dx \cdot 1) - \left(\frac{\partial P}{\partial x} dx \right) (dy \cdot 1) = \left(\frac{\partial \tau}{\partial y} - \frac{\partial P}{\partial x} \right) (dx \cdot dy \cdot 1) \\ &= \left(\mu \frac{\partial^2 u}{\partial y^2} - \frac{\partial P}{\partial x} \right) (dx \cdot dy \cdot 1) \end{aligned} \quad (6-27)$$

since $\tau = \mu(\partial u / \partial y)$. Substituting Eqs. 6-24, 6-26, and 6-27 into Eq. 6-23 and dividing by $dx \cdot dy \cdot 1$ gives

$$\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = \mu \frac{\partial^2 u}{\partial y^2} - \frac{\partial P}{\partial x} \quad (6-28)$$

This is the relation for the **momentum balance** in the x -direction, and it is known as the **x -momentum equation**. Note that we would obtain the same result if we used momentum flow rates for the left-hand side of this equation instead of mass times acceleration. If there is a body force acting in the x -direction, it can be added to the right side of the equation, provided that it is expressed per unit volume of the fluid.

In a boundary layer, the velocity component in the flow direction is much larger than that in the normal direction, and thus $u \gg v$, and $\partial v/\partial x$ and $\partial v/\partial y$ are negligible. Also, u varies greatly with y in the normal direction from zero at the wall surface to nearly the free-stream value across the relatively thin boundary layer, while the variation of u with x along the flow is typically small. Therefore, $\partial u/\partial y \gg \partial u/\partial x$. Similarly, if the fluid and the wall are at different temperatures and the fluid is heated or cooled during flow, heat conduction occurs primarily in the direction normal to the surface, and thus $\partial T/\partial y \gg \partial T/\partial x$. That is, the velocity and temperature gradients normal to the surface are much greater than those along the surface. These simplifications are known as the **boundary layer approximations**. These approximations greatly simplify the analysis, usually with little loss in accuracy, and they make it possible to obtain analytical solutions for certain types of flow problems (Fig. 6–30).

When gravity effects and other body forces are negligible and the boundary layer approximations are valid, applying Newton's second law of motion on the volume element in the y -direction gives the y -momentum equation to be

$$\frac{\partial P}{\partial y} = 0 \quad (6-29)$$

That is, *the variation of pressure in the direction normal to the surface is negligible*, and thus $P = P(x)$ and $\partial P/\partial x = dP/dx$. Then it follows that for a given x , the pressure in the boundary layer is equal to the pressure in the free stream, and the pressure determined by a separate analysis of fluid flow in the free stream (which is typically easier because of the absence of viscous effects) can readily be used in the boundary layer analysis.

The velocity components in the free stream region of a flat plate are $u = V = \text{constant}$ and $v = 0$. Substituting these into the x -momentum equations (Eq. 6–28) gives $\partial P/\partial x = 0$. Therefore, for flow over a flat plate, the pressure remains constant over the entire plate (both inside and outside the boundary layer).

Conservation of Energy Equation

The energy balance for any system undergoing any process is expressed as $E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$, which states that the change in the energy content of a system during a process is equal to the difference between the energy input and the energy output. During a *steady-flow process*, the total energy content of a control volume remains constant (and thus $\Delta E_{\text{system}} = 0$), and the amount of energy entering a control volume in all forms must be equal to the amount of energy leaving it. Then the rate form of the general energy equation reduces for a steady-flow process to $\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = 0$.

Noting that energy can be transferred by heat, work, and mass only, the energy balance for a steady-flow control volume can be written explicitly as

$$(\dot{E}_{\text{in}} - \dot{E}_{\text{out}})_{\text{by heat}} + (\dot{E}_{\text{in}} - \dot{E}_{\text{out}})_{\text{by work}} + (\dot{E}_{\text{in}} - \dot{E}_{\text{out}})_{\text{by mass}} = 0 \quad (6-30)$$

The total energy of a flowing fluid stream per unit mass is $e_{\text{stream}} = h + \text{ke} + \text{pe}$ where h is the enthalpy (which is the sum of internal energy and flow energy), $\text{pe} = gz$ is the potential energy, and $\text{ke} = V^2/2 = (u^2 + v^2)/2$ is the kinetic energy of the fluid per unit mass. The kinetic and potential energies are usually very small relative to enthalpy, and therefore it is common practice to neglect them (besides, it can be shown that if kinetic energy is included in the

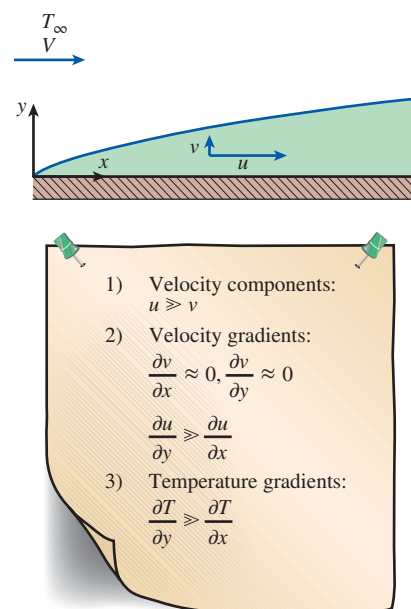
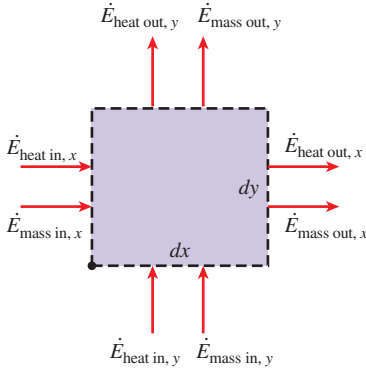


FIGURE 6–30

Boundary layer approximations.

**FIGURE 6-31**

The energy transfers by heat and mass flow associated with a differential control volume in the thermal boundary layer in steady two-dimensional flow.

following analysis, all the terms due to this inclusion cancel each other). We assume the density ρ , specific heat c_p , viscosity μ , and the thermal conductivity k of the fluid to be constant. Then the energy of the fluid per unit mass can be expressed as $e_{\text{stream}} = h = c_p T$.

Energy is a scalar quantity, and thus energy interactions in all directions can be combined in one equation. Noting that the mass flow rate of the fluid entering the control volume from the left is $\rho u(dy \cdot 1)$, the rate of energy transfer to the control volume by mass in the x -direction is, from Fig. 6–31,

$$\begin{aligned} (\dot{E}_{\text{in}} - \dot{E}_{\text{out}})_{\text{by mass}, x} &= (\dot{m}e_{\text{stream}})_x - \left[(\dot{m}e_{\text{stream}})_x + \frac{\partial(\dot{m}e_{\text{stream}})_x}{\partial x} dx \right] \\ &= -\frac{\partial[\rho u(dy \cdot 1)c_p T]}{\partial x} dx = -\rho c_p \left(u \frac{\partial T}{\partial x} + T \frac{\partial u}{\partial x} \right) dx dy \end{aligned} \quad (6-31)$$

Repeating this for the y -direction and adding the results, the net rate of energy transfer to the control volume by mass is determined to be

$$\begin{aligned} (\dot{E}_{\text{in}} - \dot{E}_{\text{out}})_{\text{by mass}} &= -\rho c_p \left(u \frac{\partial T}{\partial x} + T \frac{\partial u}{\partial x} \right) dx dy - \rho c_p \left(v \frac{\partial T}{\partial y} + T \frac{\partial v}{\partial y} \right) dx dy \\ &= -\rho c_p \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) dx dy \end{aligned} \quad (6-32)$$

since $\partial u / \partial x + \partial v / \partial y = 0$ from the continuity equation.

The net rate of heat conduction to the volume element in the x -direction is

$$\begin{aligned} (\dot{E}_{\text{in}} - \dot{E}_{\text{out}})_{\text{by heat}, x} &= \dot{Q}_x - \left(\dot{Q}_x + \frac{\partial \dot{Q}_x}{\partial x} dx \right) = -\frac{\partial}{\partial x} \left(-k(dy \cdot 1) \frac{\partial T}{\partial x} \right) dx \\ &= k \frac{\partial^2 T}{\partial x^2} dx dy \end{aligned} \quad (6-33)$$

Repeating this for the y -direction and adding the results, the net rate of energy transfer to the control volume by heat conduction becomes

$$(\dot{E}_{\text{in}} - \dot{E}_{\text{out}})_{\text{by heat}} = k \frac{\partial^2 T}{\partial x^2} dx dy + k \frac{\partial^2 T}{\partial y^2} dx dy = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) dx dy \quad (6-34)$$

Another mechanism of energy transfer to and from the fluid in the control volume is the work done by the body and surface forces. The work done by a body force is determined by multiplying this force by the velocity in the direction of the force and the volume of the fluid element, and this work needs to be considered only in the presence of significant gravitational, electric, or magnetic effects. The surface forces consist of the forces due to fluid pressure and the viscous shear stresses. The work done by pressure (the flow work) is already accounted for in the analysis above by using enthalpy for the microscopic energy of the fluid instead of internal energy. The shear stresses that result from viscous effects are usually very small and can be neglected in many cases. This is especially the case for applications that involve low or moderate velocities.

Then the energy equation for the steady two-dimensional flow of a fluid with constant properties and negligible shear stresses is obtained by substituting Eqs. 6–32 and 6–34 into 6–30 to be

$$\rho c_p \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) \quad (6-35)$$

which states that *the net energy convected by the fluid out of the control volume is equal to the net energy transferred into the control volume by heat conduction*.

When the viscous shear stresses are not negligible, their effect is accounted for by expressing the energy equation as

$$\rho c_p \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) + \mu \Phi \quad (6-36)$$

where the *viscous dissipation function* Φ is obtained after a lengthy analysis [see an advanced book such as the one by *Schlichting* (1979) for details] to be

$$\Phi = 2 \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 \right] + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)^2 \quad (6-37)$$

Viscous dissipation may play a dominant role in high-speed flows, especially when the viscosity of the fluid is high (like the flow of oil in journal bearings). This manifests itself as a significant rise in fluid temperature due to the conversion of the kinetic energy of the fluid to thermal energy. Viscous dissipation is also significant for high-speed flights of aircraft.

For the special case of a stationary fluid, $u = v = 0$, the energy equation reduces, as expected, to the steady two-dimensional heat conduction equation,

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0 \quad (6-38)$$

EXAMPLE 6-2 Temperature Rise of Oil in a Journal Bearing

The flow of oil in a journal bearing can be approximated as parallel flow between two large plates with one plate moving and the other stationary. Such flows are known as Couette flow.

Consider two large isothermal plates separated by 2-mm-thick oil film. The upper plate moves at a constant velocity of 12 m/s, while the lower plate is stationary. Both plates are maintained at 20°C. (a) Obtain relations for the velocity and temperature distributions in the oil. (b) Determine the maximum temperature in the oil and the heat flux from the oil to each plate (Fig. 6-32).

SOLUTION Parallel flow of oil between two plates is considered. The velocity and temperature distributions, the maximum temperature, and the total heat transfer rate are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Oil is an incompressible substance with constant properties. 3 Body forces such as gravity are negligible. 4 The plates are large, so there is no variation in the z -direction.

Properties The properties of oil at 20°C are (Table A-13):

$$k = 0.145 \text{ W/m}\cdot\text{K} \quad \text{and} \quad \mu = 0.8374 \text{ kg/m}\cdot\text{s} = 0.8374 \text{ N}\cdot\text{s/m}^2$$

Analysis (a) We take the x -axis to be the flow direction and y to be the normal direction. This is parallel flow between two plates, and thus $v = 0$. Then the continuity equation (Eq. 6-21) reduces to

$$\text{Continuity:} \quad \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \rightarrow \frac{\partial u}{\partial x} = 0 \rightarrow u = u(y)$$

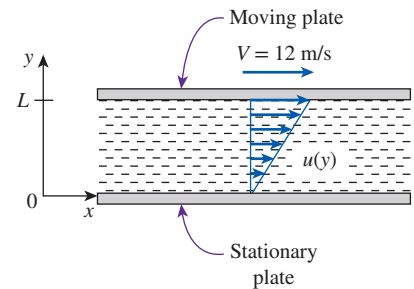


FIGURE 6-32
Schematic for Example 6-2.

Therefore, the x -component of velocity does not change in the flow direction (i.e., the velocity profile remains unchanged). Noting that $u = u(y)$, $v = 0$, and $\partial P/\partial x = 0$ (flow is maintained by the motion of the upper plate rather than the pressure gradient), the x -momentum equation (Eq. 6–28) reduces to

$$x\text{-momentum:} \quad \rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = \mu \frac{\partial^2 u}{\partial y^2} - \frac{\partial P}{\partial x} \rightarrow \frac{d^2 u}{dy^2} = 0$$

This is a second-order ordinary differential equation, and integrating it twice gives

$$u(y) = C_1 y + C_2$$

The fluid velocities at the plate surfaces must be equal to the velocities of the plates because of the no-slip condition. Therefore, the boundary conditions are $u(0) = 0$ and $u(L) = V$, and applying them gives the velocity distribution to be

$$u(y) = \frac{y}{L} V$$

Frictional heating due to viscous dissipation in this case is significant because of the high viscosity of oil and the large plate velocity. The plates are isothermal, and there is no change in the flow direction, and thus the temperature depends on y only, $T = T(y)$. Also, $u = u(y)$ and $v = 0$. Then the energy equation with dissipation (Eqs. 6–36 and 6–37) reduce to

$$Energy: \quad 0 = k \frac{\partial^2 T}{\partial y^2} + \mu \left(\frac{\partial u}{\partial y} \right)^2 \rightarrow k \frac{d^2 T}{dy^2} = -\mu \left(\frac{V}{L} \right)^2$$

since $\partial u/\partial y = V/L$. Dividing both sides by k and integrating twice give

$$T(y) = -\frac{\mu}{2k} \left(\frac{y}{L} V \right)^2 + C_3 y + C_4$$

Applying the boundary conditions $T(0) = T_0$ and $T(L) = T_0$ gives the temperature distribution to be

$$T(y) = T_0 + \frac{\mu V^2}{2k} \left(\frac{y}{L} - \frac{y^2}{L^2} \right)$$

(b) The temperature gradient is determined by differentiating $T(y)$ with respect to y ,

$$\frac{dT}{dy} = \frac{\mu V^2}{2kL} \left(1 - 2 \frac{y}{L} \right)$$

The location of maximum temperature is determined by setting $dT/dy = 0$ and solving for y ,

$$\frac{dT}{dy} = \frac{\mu V^2}{2kL} \left(1 - 2 \frac{y}{L} \right) = 0 \rightarrow y = \frac{L}{2}$$

Therefore, maximum temperature occurs at midplane, which is not surprising since both plates are maintained at the same temperature. The maximum temperature is the value of temperature at $y = L/2$,

$$\begin{aligned} T_{\max} &= T\left(\frac{L}{2}\right) = T_0 + \frac{\mu V^2}{2k} \left(\frac{L/2}{L} - \frac{(L/2)^2}{L^2} \right) = T_0 + \frac{\mu V^2}{8k} \\ &= 20 + \frac{(0.8374 \text{ N}\cdot\text{s/m}^2)(12 \text{ m/s})^2}{8(0.145 \text{ W/m}\cdot\text{K})} \left(\frac{1 \text{ W}}{1 \text{ N}\cdot\text{m/s}} \right) = 124^\circ\text{C} \end{aligned}$$

Heat flux at the plates is determined from the definition of heat flux,

$$\begin{aligned}\dot{q}_0 &= -k \left. \frac{dT}{dy} \right|_{y=0} = -k \frac{\mu V^2}{2kL} (1 - 0) = -\frac{\mu V^2}{2L} \\ &= -\frac{(0.8374 \text{ N}\cdot\text{s/m}^2)(12 \text{ m/s})^2}{2(0.002 \text{ m})} \left(\frac{1 \text{ kW}}{1000 \text{ N}\cdot\text{m/s}} \right) = -30.1 \text{ kW/m}^2 \\ \dot{q}_L &= -k \left. \frac{dT}{dy} \right|_{y=L} = -k \frac{\mu V^2}{2kL} (1 - 2) = \frac{\mu V^2}{2L} = -\dot{q}_0 = 30.1 \text{ kW/m}^2\end{aligned}$$

Therefore, heat fluxes at the two plates are equal in magnitude but opposite in sign.

Discussion A temperature rise of 104°C confirms our suspicion that viscous dissipation is very significant. Also, the heat flux is equivalent to the rate of mechanical energy dissipation. Therefore, mechanical energy is being converted to thermal energy at a rate of 60.2 kW/m² of plate area to overcome friction in the oil. Finally, calculations are done using oil properties at 20°C, but the oil temperature turned out to be much higher. Therefore, knowing the strong dependence of viscosity on temperature, calculations should be repeated using properties at the average temperature of 72°C to improve accuracy.

6-8 ■ SOLUTIONS OF CONVECTION EQUATIONS FOR A FLAT PLATE

Consider laminar flow of a fluid over a *flat plate*, as shown in Fig. 6–33. Surfaces that are slightly contoured such as turbine blades can also be approximated as flat plates with reasonable accuracy. The x -coordinate is measured along the plate surface from the leading edge of the plate in the direction of the flow, and y is measured from the surface in the normal direction. The fluid approaches the plate in the x -direction with a uniform upstream velocity, which is equivalent to the free stream velocity V .

When viscous dissipation is negligible, the continuity, momentum, and energy equations (Eqs. 6–21, 6–28, and 6–35) reduce for steady, incompressible, laminar flow of a fluid with constant properties over a flat plate to

$$\text{Continuity:} \quad \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (6-39)$$

$$\text{Momentum:} \quad u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial y^2} \quad (6-40)$$

$$\text{Energy:} \quad u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2} \quad (6-41)$$

with the boundary conditions (Fig. 6–25)

$$\begin{aligned}\text{At } x = 0: & \quad u(0, y) = V, \quad T(0, y) = T_\infty \\ \text{At } y = 0: & \quad u(x, 0) = 0, \quad v(x, 0) = 0, \quad T(x, 0) = T_w \\ \text{As } y \rightarrow \infty: & \quad u(x, \infty) = V, \quad T(x, \infty) = T_\infty\end{aligned} \quad (6-42)$$

When fluid properties are assumed to be constant and thus independent of temperature, the first two equations can be solved separately for the velocity

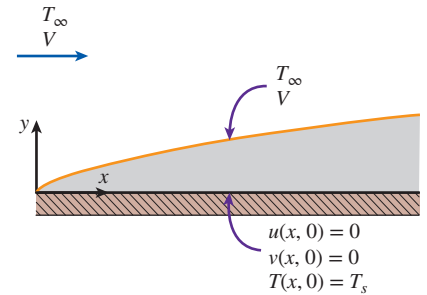


FIGURE 6-33

Boundary conditions for flow over a flat plate.

components u and v . Once the velocity distribution is available, we can determine the friction coefficient and the boundary layer thickness using their definitions. Also, knowing u and v , the temperature becomes the only unknown in the last equation, and it can be solved for temperature distribution.

The continuity and momentum equations were first solved in 1908 by the German engineer H. Blasius, a student of L. Prandtl. This was done by transforming the two partial differential equations into a single ordinary differential equation by introducing a new independent variable, called the **similarity variable**. The finding of such a variable, assuming it exists, is more of an art than science, and it requires to have a good insight into the problem.

Noticing that the general shape of the velocity profile remains the same along the plate, Blasius reasoned that the nondimensional velocity profile u/V should remain unchanged when plotted against the nondimensional distance y/δ , where δ is the thickness of the local velocity boundary layer at a given x . That is, although both δ and u at a given y vary with x , the velocity u at a fixed y/δ remains constant. Blasius was also aware from the work of Stokes that δ is proportional to $\sqrt{\nu x/V}$, and thus he defined a *dimensionless similarity variable* as

$$\eta = y \sqrt{\frac{V}{\nu x}} \quad (6-43)$$

and thus $u/V = \text{function}(\eta)$. He then introduced a *stream function* $\psi(x, y)$ as

$$u = \frac{\partial \psi}{\partial y} \quad \text{and} \quad v = -\frac{\partial \psi}{\partial x} \quad (6-44)$$

so that the continuity equation (Eq. 6-39) is automatically satisfied and thus eliminated (this can be verified easily by direct substitution). Next he defined a function $f(\eta)$ as the dependent variable as

$$f(\eta) = \frac{\psi}{V \sqrt{\nu x/V}} \quad (6-45)$$

Then the velocity components become

$$u = \frac{\partial \psi}{\partial y} = \frac{\partial \psi}{\partial \eta} \frac{\partial \eta}{\partial y} = V \sqrt{\frac{\nu x}{V}} \frac{df}{d\eta} \sqrt{\frac{V}{\nu x}} = V \frac{df}{d\eta} \quad (6-46)$$

$$v = -\frac{\partial \psi}{\partial x} = -V \sqrt{\frac{\nu x}{V}} \frac{df}{d\eta} - \frac{V}{2} \sqrt{\frac{\nu}{Vx}} f = \frac{1}{2} \sqrt{\frac{V\nu}{x}} \left(\eta \frac{df}{d\eta} - f \right) \quad (6-47)$$

By differentiating these u and v relations, the derivatives of the velocity components can be shown to be

$$\frac{\partial u}{\partial x} = -\frac{V}{2x} \eta \frac{d^2 f}{d\eta^2}, \quad \frac{\partial u}{\partial y} = V \sqrt{\frac{V}{\nu x}} \frac{d^2 f}{d\eta^2}, \quad \frac{\partial^2 u}{\partial y^2} = \frac{V^2}{\nu x} \frac{d^3 f}{d\eta^3} \quad (6-48)$$

Substituting these relations into the momentum equation and simplifying, we obtain

$$2 \frac{d^3 f}{d\eta^3} + f \frac{d^2 f}{d\eta^2} = 0 \quad (6-49)$$

which is a third-order nonlinear differential equation. Therefore, the system of two partial differential equations is transformed into a single ordinary differential equation by the use of a similarity variable. Using the definitions of

f and η , the boundary conditions in terms of the similarity variables can be expressed as

$$f(0) = 0, \quad \left. \frac{df}{d\eta} \right|_{\eta=0} = 0, \quad \text{and} \quad \left. \frac{df}{d\eta} \right|_{\eta=\infty} = 1 \quad (6-50)$$

The transformed equation with its associated boundary conditions cannot be solved analytically, and thus an alternative solution method is necessary. The problem was first solved by Blasius in 1908 using a power series expansion approach, and this original solution is known as the *Blasius solution*. The problem is later solved more accurately using different numerical approaches, and results from such a solution are given in Table 6–3. The nondimensional velocity profile can be obtained by plotting u/V against η . The results obtained by this simplified analysis are in excellent agreement with experimental results.

Recall that we defined the boundary layer thickness as the distance from the surface for which $u/V = 0.99$. We observe from Table 6–3 that the value of η corresponding to $u/V = 0.99$ is $\eta = 4.91$. Substituting $\eta = 4.91$ and $y = \delta$ into the definition of the similarity variable (Eq. 6–43) gives $4.91 = \delta \sqrt{V/\nu x}$. Then the velocity boundary layer thickness becomes

$$\delta = \frac{4.91}{\sqrt{V/\nu x}} = \frac{4.91x}{\sqrt{\text{Re}_x}} \quad (6-51)$$

since $\text{Re}_x = Vx/\nu$, where x is the distance from the leading edge of the plate. Note that the boundary layer thickness increases with increasing kinematic viscosity ν and with increasing distance from the leading edge x , but it decreases with increasing free-stream velocity V . Therefore, a large free-stream velocity suppresses the boundary layer and causes it to be thinner.

The shear stress on the wall can be determined from its definition and the $\partial u/\partial y$ relation in Eq. 6–48:

$$\tau_w = \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} = \mu V \sqrt{\frac{V}{\nu x}} \left. \frac{d^2 f}{d\eta^2} \right|_{\eta=0} \quad (6-52)$$

Substituting the value of the second derivative of f at $\eta = 0$ from Table 6–3 gives

$$\tau_w = 0.332V \sqrt{\frac{\rho \mu V}{x}} = \frac{0.332\rho V^2}{\sqrt{\text{Re}_x}} \quad (6-53)$$

Then the local friction coefficient becomes

$$C_{f,x} = \frac{\tau_w}{\rho V^2/2} = 0.664 \text{Re}_x^{-1/2} \quad (6-54)$$

Note that unlike the boundary layer thickness, wall shear stress and the skin friction coefficient decrease along the plate as $x^{-1/2}$.

The Energy Equation

Knowing the velocity profile, we are now ready to solve the energy equation for temperature distribution for the case of constant wall temperature T_s . First we introduce the dimensionless temperature θ as

$$\theta(x, y) = \frac{T(x, y) - T_s}{T_\infty - T_s} \quad (6-55)$$

TABLE 6–3

Similarity function f and its derivatives for laminar boundary layer along a flat plate

η	f	$\frac{df}{d\eta} = \frac{u}{V}$	$\frac{d^2 f}{d\eta^2}$
0	0	0	0.332
0.5	0.042	0.166	0.331
1.0	0.166	0.330	0.323
1.5	0.370	0.487	0.303
2.0	0.650	0.630	0.267
2.5	0.996	0.751	0.217
3.0	1.397	0.846	0.161
3.5	1.838	0.913	0.108
4.0	2.306	0.956	0.064
4.5	2.790	0.980	0.034
5.0	3.283	0.992	0.016
5.5	3.781	0.997	0.007
6.0	4.280	0.999	0.002
∞	∞	1	0

Noting that both T_s and T_∞ are constant, substitution into the energy equation Eq. 6-41 gives

$$u \frac{\partial \theta}{\partial x} + v \frac{\partial \theta}{\partial y} = \alpha \frac{\partial^2 \theta}{\partial y^2} \quad (6-56)$$

Temperature profiles for flow over an isothermal flat plate are similar, just like the velocity profiles, and thus we expect a similarity solution for temperature to exist. Further, the thickness of the thermal boundary layer is proportional to $\sqrt{\nu x/V}$, just like the thickness of the velocity boundary layer, and thus the similarity variable is also η , and $\theta = \theta(\eta)$. Using the chain rule and substituting the u and v expressions from Eqs. 6-46 and 6-47 into the energy equation gives

$$V \frac{df}{d\eta} \frac{d\theta}{d\eta} \frac{\partial \eta}{\partial x} + \frac{1}{2} \sqrt{\frac{V y}{x}} \left(\eta \frac{df}{d\eta} \right) \frac{d\theta}{d\eta} \frac{\partial \eta}{\partial y} = \alpha \frac{d^2 \theta}{d\eta^2} \left(\frac{\partial \eta}{\partial y} \right)^2 \quad (6-57)$$

Simplifying and noting that $\text{Pr} = \nu/\alpha$ gives

$$2 \frac{d^2 \theta}{d\eta^2} + \text{Pr} f \frac{d\theta}{d\eta} = 0 \quad (6-58)$$

with the boundary conditions $\theta(0) = 0$ and $\theta(\infty) = 1$. Obtaining an equation for θ as a function of η alone confirms that the temperature profiles are similar, and thus a similarity solution exists. Again a closed-form solution cannot be obtained for this boundary value problem, and it must be solved numerically.

It is interesting to note that for $\text{Pr} = 1$, this equation reduces to Eq. 6-49 when θ is replaced by $df/d\eta$, which is equivalent to u/V (see Eq. 6-46). The boundary conditions for θ and $df/d\eta$ are also identical. Thus we conclude that the velocity and thermal boundary layers coincide, and the nondimensional velocity and temperature profiles (u/V and θ) are identical for steady, incompressible, laminar flow of a fluid with constant properties and $\text{Pr} = 1$ over an isothermal flat plate (Fig. 6-34). The value of the temperature gradient at the surface ($y = 0$ or $\eta = 0$) in this case is, from Table 6-3, $d\theta/d\eta = d^2 f/d\eta^2 = 0.332$.

Equation 6-58 is solved for numerous values of Prandtl numbers. For $\text{Pr} > 0.6$, the nondimensional temperature gradient at the surface is found to be proportional to $\text{Pr}^{1/3}$ and is expressed as

$$\left. \frac{d\theta}{d\eta} \right|_{\eta=0} = 0.332 \text{Pr}^{1/3} \quad (6-59)$$

The temperature gradient at the surface is

$$\begin{aligned} \left. \frac{\partial T}{\partial y} \right|_{y=0} &= (T_\infty - T_s) \left. \frac{\partial \theta}{\partial y} \right|_{y=0} = (T_\infty - T_s) \left. \frac{d\theta}{d\eta} \right|_{\eta=0} \left. \frac{\partial \eta}{\partial y} \right|_{y=0} \\ &= 0.332 \text{Pr}^{1/3} (T_\infty - T_s) \sqrt{\frac{V}{\nu x}} \end{aligned} \quad (6-60)$$

Then the local convection coefficient and Nusselt number become

$$h_x = \frac{\dot{q}_s}{T_s - T_\infty} = \frac{-k(\partial T/\partial y)|_{y=0}}{T_s - T_\infty} = 0.332 \text{Pr}^{1/3} k \sqrt{\frac{V}{\nu x}} \quad (6-61)$$

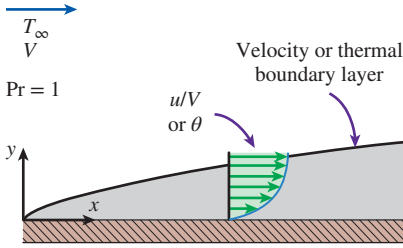


FIGURE 6-34

When $\text{Pr} = 1$, the velocity and thermal boundary layers coincide, and the nondimensional velocity and temperature profiles are identical for steady, incompressible, laminar flow over a flat plate.

and

$$\text{Nu}_x = \frac{h_x x}{k} = 0.332 \text{Pr}^{1/3} \text{Re}_x^{1/2} \quad \text{Pr} > 0.6 \quad (6-62)$$

The Nu_x values obtained from this relation agree well with measured values.

Solving Eq. 6-58 numerically for the temperature profile for different Prandtl numbers, and using the definition of the thermal boundary layer, it is determined that $\delta/\delta_t \cong \text{Pr}^{1/3}$. Then the thermal boundary layer thickness becomes

$$\delta_t = \frac{\delta}{\text{Pr}^{1/3}} = \frac{4.91x}{\text{Pr}^{1/3} \sqrt{\text{Re}_x}} \quad (6-63)$$

Note that these relations are valid only for laminar flow over an isothermal flat plate. Also, the effect of variable properties can be accounted for by evaluating all such properties at the film temperature defined as $T_f = (T_s + T_\infty)/2$.

The Blasius solution gives important insights, but its value is largely historical because of the limitations it involves. Today both laminar and turbulent flows over surfaces are routinely analyzed using numerical methods.

6-9 ■ NONDIMENSIONALIZED CONVECTION EQUATIONS AND SIMILARITY

When viscous dissipation is negligible, the continuity, momentum, and energy equations for steady, laminar flow of a fluid with constant properties are given by Eqs. 6-21, 6-28, and 6-35.

These equations and the boundary conditions can be nondimensionalized by dividing all dependent and independent variables by relevant and meaningful constant quantities: all lengths by a characteristic length L (which is the length for a plate), all velocities by a reference velocity V (which is the free stream velocity for a plate), pressure by ρV^2 (which is twice the free stream dynamic pressure for a plate), and temperature by a suitable temperature difference (which is $T_\infty - T_s$ for a plate). We get

$$x^* = \frac{x}{L}, \quad y^* = \frac{y}{L}, \quad u^* = \frac{u}{V}, \quad v^* = \frac{v}{V}, \quad P^* = \frac{P}{\rho V^2} \quad \text{and} \quad T^* = \frac{T - T_s}{T_\infty - T_s}$$

where the asterisks are used to denote nondimensional variables. Introducing these variables into Eqs. 6-21, 6-28, and 6-41 and simplifying give

$$\text{Continuity:} \quad \frac{\partial u^*}{\partial x^*} + \frac{\partial v^*}{\partial y^*} = 0 \quad (6-64)$$

$$\text{Momentum:} \quad u^* \frac{\partial u^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} = \frac{1}{\text{Re}_L} \frac{\partial^2 u^*}{\partial y^{*2}} - \frac{dP^*}{dx^*} \quad (6-65)$$

$$\text{Energy:} \quad u^* \frac{\partial T^*}{\partial x^*} + v^* \frac{\partial T^*}{\partial y^*} = \frac{1}{\text{Re}_L \text{Pr}} \frac{\partial^2 T^*}{\partial y^{*2}} \quad (6-66)$$

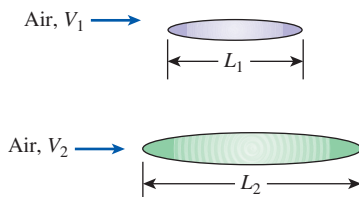
with the boundary conditions

$$\begin{aligned} u^*(0, y^*) &= 1, & u^*(x^*, 0) &= 0, & u^*(x^*, \infty) &= 1, & v^*(x^*, 0) &= 0, \\ T^*(0, y^*) &= 1, & T^*(x^*, 0) &= 0, & T^*(x^*, \infty) &= 1 \end{aligned} \quad (6-67)$$

Parameters before nondimensionalizing:
 $L, V, T_\infty, T_s, \nu, \alpha$
 Parameters after nondimensionalizing:
 Re, Pr

FIGURE 6–35

The number of parameters is reduced greatly by nondimensionalizing the convection equations.

**FIGURE 6–36**

Schematic for Example 6–3.

where $Re_L = VL/\nu$ is the dimensionless Reynolds number and $Pr = \nu/\alpha$ is the Prandtl number. For a given type of geometry, the solutions to problems with the same Re and Pr numbers are similar, and thus Re and Pr numbers serve as *similarity parameters*. Two physical phenomena are *similar* if they have the same dimensionless forms of governing differential equations and boundary conditions (see Example 6–3).

A major advantage of nondimensionalizing is the significant reduction in the number of parameters. The original problem involves six parameters ($L, V, T_\infty, T_s, \nu, \alpha$), but the nondimensionalized problem involves just two parameters (Re_L and Pr). For a given geometry, problems that have the same values for the similarity parameters have identical solutions. For example, determining the convection heat transfer coefficient for flow over a given surface requires numerical solutions or experimental investigations for several fluids, with several sets of velocities, surface lengths, wall temperatures, and free stream temperatures. The same information can be obtained with far fewer investigations by grouping data into the dimensionless Re and Pr numbers. Another advantage of similarity parameters is that they enable us to group the results of a large number of experiments and to report them conveniently in terms of such parameters (Fig. 6–35).

EXAMPLE 6–3 Finding Rate of Heat Transfer Using Similarity Parameters

An airfoil with a characteristic length of 0.15 m and a surface temperature of 325°C is placed in airflow at 37°C with free stream velocity of 100 m/s. The rate of heat transfer from the airfoil is 1600 W. If a second airfoil with a characteristic length of 0.3 m and a surface temperature of 425°C is placed in the airflow at 37°C with free stream velocity of 50 m/s, determine the rate of heat transfer from the second airfoil. The surface area of the airfoil may be assumed to be directly proportional to its characteristic length (Fig. 6–36).

SOLUTION Two airfoils with different characteristic lengths and surface temperatures are placed in airflow of different free stream velocities at 37°C. The rate of heat transfer from the second airfoil is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Properties are constant. 3 Both airfoils are geometrically similar. 4 Surface area A is directly proportional to characteristic length L .

Analysis The relation for Nusselt, Prandtl, and Reynolds numbers is given as

$$Nu = g(Re, Pr)$$

$$\text{where } Nu = \frac{hL}{k}, \quad Pr = \frac{c_p \mu}{k}, \quad Re = \frac{VL}{\nu}$$

Then, for airfoils 1 and 2 we have

$$Re_1 = \frac{V_1 L_1}{\nu} = \frac{(100 \text{ m/s})(0.15 \text{ m})}{\nu} = \frac{15 \text{ m}^2/\text{s}}{\nu} \quad \text{and} \quad Pr_1 = \frac{c_p \mu}{k}$$

$$Re_2 = \frac{V_2 L_2}{\nu} = \frac{(50 \text{ m/s})(0.3 \text{ m})}{\nu} = \frac{15 \text{ m}^2/\text{s}}{\nu} \quad \text{and} \quad Pr_2 = \frac{c_p \mu}{k}$$

Since the fluid properties are constant, we have $Re_1 = Re_2$ and $Pr_1 = Pr_2$, which implies

$$Nu_1 = g(Re_1, Pr_1) = Nu_2 = g(Re_2, Pr_2) \rightarrow Nu_1 = Nu_2$$

Hence

$$\frac{h_1 L_1}{k} = \frac{h_2 L_2}{k} \rightarrow h_2 = h_1 \frac{L_1}{L_2} = \frac{\dot{Q}_1}{A_1 (T_{s,1} - T_\infty)} \frac{L_1}{L_2}$$

$$\dot{Q}_2 = h_2 A_2 (T_{s,2} - T_\infty) = \frac{L_1 A_2 (T_{s,2} - T_\infty)}{L_2 A_1 (T_{s,1} - T_\infty)} \dot{Q}_1$$

$$\dot{Q}_2 = \frac{(T_{s,2} - T_\infty)}{(T_{s,1} - T_\infty)} \dot{Q}_1 = \frac{(425 - 37)^\circ\text{C}}{(325 - 37)^\circ\text{C}} (1600 \text{ W}) = \mathbf{2156 \text{ W}}$$

Discussion The relation for the Nusselt numbers $Nu_1 = Nu_2$ is valid due to $Re_1 = Re_2$ and $Pr_1 = Pr_2$. Slight variation of ν from airfoil 1 to airfoil 2 would cause Re_2 to differ from Re_1 . However, for the given conditions, the variable property effect is small.

6-10 ■ FUNCTIONAL FORMS OF FRICTION AND CONVECTION COEFFICIENTS

The three nondimensionalized boundary layer equations (Eqs. 6-64, 6-65, and 6-66) involve three unknown functions u^* , v^* , and T^* , two independent variables x^* and y^* , and two parameters Re_L and Pr . The pressure $P^*(x^*)$ depends on the geometry involved (it is constant for a flat plate), and it has the same value inside and outside the boundary layer at a specified x^* . Therefore, it can be determined separately from the free stream conditions, and dP^*/dx^* in Eq. 6-65 can be treated as a known function of x^* . Note that the boundary conditions do not introduce any new parameters.

For a given geometry, the solution for u^* can be expressed as

$$u^* = f_1(x^*, y^*, Re_L) \quad (6-68)$$

Then the shear stress at the surface becomes

$$\tau_w = \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} = \frac{\mu V}{L} \left. \frac{\partial u^*}{\partial y^*} \right|_{y^*=0} = \frac{\mu V}{L} f_2(x^*, Re_L) \quad (6-69)$$

Substituting into its definition gives the local friction coefficient,

$$C_{f,x} = \frac{\tau_w}{\rho V^2/2} = \frac{\mu V/L}{\rho V^2/2} f_2(x^*, Re_L) = \frac{2}{Re_L} f_2(x^*, Re_L) = f_3(x^*, Re_L) \quad (6-70)$$

Thus we conclude that the friction coefficient for a given geometry can be expressed in terms of the Reynolds number Re and the dimensionless space variable x^* alone (instead of being expressed in terms of x , L , V , ρ , and μ). This is a very significant finding, and it shows the value of nondimensionalized equations.

Similarly, the solution of Eq. 6–66 for the dimensionless temperature T^* for a given geometry can be expressed as

$$T^* = g_1(x^*, y^*, \text{Re}_L, \text{Pr}) \quad (6-71)$$

Using the definition of T^* , the convection heat transfer coefficient becomes

$$h_x = \frac{-k(\partial T/\partial y)|_{y=0}}{T_s - T_\infty} = \frac{-k(T_\infty - T_s)}{L(T_s - T_\infty)} \frac{\partial T^*}{\partial y^*} \bigg|_{y^*=0} = \frac{k}{L} \frac{\partial T^*}{\partial y^*} \bigg|_{y^*=0} \quad (6-72)$$

Substituting this into the Nusselt number relation gives [or alternately, we can rearrange the relation above in dimensionless form as $hL/k = (\partial T^*/\partial y^*)|_{y^*=0}$ and define the dimensionless group hL/k as the Nusselt number]

$$\text{Nu}_x = \frac{h_x x}{k} = \frac{\partial T^*}{\partial y^*} \bigg|_{y^*=0} = g_2(x^*, \text{Re}_x, \text{Pr}) \quad (6-73)$$

Note that the Nusselt number is equivalent to the *dimensionless temperature gradient at the surface*, and thus it is properly referred to as the dimensionless heat transfer coefficient (Fig. 6–37). Also, the Nusselt number for a given geometry can be expressed in terms of the Reynolds number Re , the Prandtl number Pr , and the space variable x^* , and such a relation can be used for different fluids flowing at different velocities over similar geometries of different lengths.

The average friction and heat transfer coefficients are determined by integrating $C_{f,x}$ and Nu_x over the surface of the given body with respect to x^* from 0 to 1. Integration removes the dependence on x^* , and the average friction coefficient and Nusselt number can be expressed as

$$C_f = f_4(\text{Re}_L) \quad \text{and} \quad \text{Nu} = g_3(\text{Re}_L, \text{Pr}) \quad (6-74)$$

These relations are extremely valuable as they state that for a given geometry, the friction coefficient can be expressed as a function of Reynolds number alone, and the Nusselt number as a function of Reynolds and Prandtl numbers alone (Fig. 6–38). Therefore, experimentalists can study a problem with a minimum number of experiments, and report their friction and heat transfer coefficient measurements conveniently in terms of Reynolds and Prandtl numbers. For example, a friction coefficient relation obtained with air for a given surface can also be used for water at the same Reynolds number. But it should be kept in mind that the validity of these relations is limited by the limitations on the boundary layer equations used in the analysis.

The experimental data for heat transfer is often represented with reasonable accuracy by a simple power-law relation of the form

$$\text{Nu} = C \text{Re}_L^m \text{Pr}^n \quad (6-75)$$

where m and n are constant exponents (usually between 0 and 1), and the value of the constant C depends on geometry. Sometimes more complex relations are used for better accuracy.

6–11 ■ ANALOGIES BETWEEN MOMENTUM AND HEAT TRANSFER

In forced convection analysis, we are primarily interested in the determination of the quantities C_f (to calculate shear stress at the wall) and Nu (to calculate heat transfer rates). Therefore, it is very desirable to have a relation between

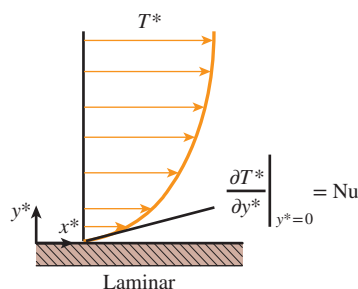


FIGURE 6–37

The Nusselt number is equivalent to the dimensionless temperature gradient at the surface.

Local Nusselt number:

$$\text{Nu}_x = \text{function}(x^*, \text{Re}_x, \text{Pr})$$

Average Nusselt number:

$$\text{Nu} = \text{function}(\text{Re}_L, \text{Pr})$$

A common form of Nusselt number:

$$\text{Nu} = C \text{Re}_L^m \text{Pr}^n$$

FIGURE 6–38

For a given geometry, the average Nusselt number is a function of Reynolds and Prandtl numbers.

C_f and Nu so that we can calculate one when the other is available. Such relations are developed on the basis of the similarity between momentum and heat transfers in boundary layers and are known as the *Reynolds analogy* (Fig. 6–20) and the *Chilton–Colburn analogy*.

Reconsider the nondimensionalized momentum and energy equations for steady, incompressible, laminar flow of a fluid with constant properties and negligible viscous dissipation (Eqs. 6–65 and 6–66). When $Pr = 1$ (which is approximately the case for gases) and $\partial P^*/\partial x^* = 0$ (which is the case when, $u = V = \text{constant}$ in the free stream, as in flow over a flat plate), these equations simplify to

$$\text{Momentum:} \quad u^* \frac{\partial u^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} = \frac{1}{Re_L} \frac{\partial^2 u^*}{\partial y^{*2}} \quad (6-76)$$

$$\text{Energy:} \quad u^* \frac{\partial T^*}{\partial x^*} + v^* \frac{\partial T^*}{\partial y^*} = \frac{1}{Re_L} \frac{\partial^2 T^*}{\partial y^{*2}} \quad (6-77)$$

which are exactly of the same form for the dimensionless velocity u^* and temperature T^* . The boundary conditions for u^* and T^* are also identical. Therefore, the functions u^* and T^* must be identical, and thus the first derivatives of u^* and T^* at the surface must be equal to each other,

$$\left. \frac{\partial u^*}{\partial y^*} \right|_{y^*=0} = \left. \frac{\partial T^*}{\partial y^*} \right|_{y^*=0} \quad (6-78)$$

Then from Eqs. 6–69, 6–70, and 6–73 we have

$$C_{f,x} \frac{Re_x}{2} = Nu_x \quad (Pr = 1) \quad (6-79)$$

which is known as the **Reynolds analogy** (Fig. 6–39). This is an important analogy since it allows us to determine the heat transfer coefficient for fluids with $Pr \approx 1$ from a knowledge of the friction coefficient, which is easier to measure. The Reynolds analogy is also expressed alternately as

$$\frac{C_{f,x}}{2} = St_x \quad (Pr = 1) \quad (6-80)$$

where

$$St = \frac{h}{\rho c_p V} = \frac{Nu}{Re_L Pr} \quad (6-81)$$

which is the **Stanton number** (Fig. 6–40), which is also a dimensionless heat transfer coefficient.

Reynolds analogy is of limited use because of the restrictions $Pr = 1$ and $\partial P^*/\partial x^* = 0$ on it, and it is desirable to have an analogy that is applicable over a wide range of Pr . This is done by adding a Prandtl number correction.

The friction coefficient and Nusselt number for a flat plate were determined in Section 6–8 to be

$$C_{f,x} = 0.664 Re_x^{-1/2} \quad \text{and} \quad Nu_x = 0.332 Pr^{1/3} Re_x^{1/2} \quad (6-82)$$

Taking their ratio and rearranging give the desired relation, known as the **modified Reynolds analogy** or **Chilton–Colburn analogy**,

$$C_{f,x} \frac{Re_x}{2} = Nu_x Pr^{-1/3} \quad \text{or} \quad \frac{C_{f,x}}{2} = St_x Pr^{2/3} = j_H \quad (6-83)$$

Profiles:	$u^* = T^*$
Gradients:	$\left. \frac{\partial u^*}{\partial y^*} \right _{y^*=0} = \left. \frac{\partial T^*}{\partial y^*} \right _{y^*=0}$
Analogy:	$C_{f,x} \frac{Re_x}{2} = Nu_x$

FIGURE 6–39

When $Pr = 1$ and $\partial P^*/\partial x^* \approx 0$, the nondimensional velocity and temperature profiles become identical, and Nu is related to C_f by the Reynolds analogy.

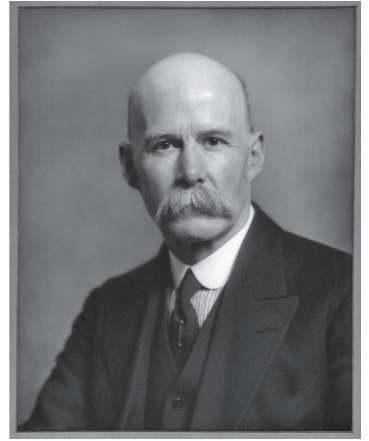


FIGURE 6–40

Sir Thomas Edward Stanton

(1865–1931), was a British engineer, born at Atherstone in Warwickshire, England. From 1891 to 1896 he worked in Osborne Reynolds' laboratory at Owens College, Manchester, England. Stanton's main field of interest was fluid flow and friction and the related problem of heat transmission. From 1902 to 1907 he executed a large research program concerning wind forces on structures, such as bridges and roofs. After 1908, the year when the Wright Brothers made their first airplane flights in Europe, Stanton was devoted to problems of airplane and airship design and the dissipation of heat from air-cooled engines. The dimensionless heat transfer coefficient **Stanton number** is named after him.

Courtesy of University of Bristol, Faculty of Engineering



FIGURE 6-41

Allan Philip Colburn (1904–1955), an American engineer, was born in Madison, Wisconsin. His research was on condensation of water vapor from saturated airstreams. He brought together for the first time in American engineering work the fundamentals of momentum, heat and mass transfer, and thermodynamic principles to deal with this complex problem. The dimensionless empirical parameter **Colburn j-factor** ($j_H = St_x Pr^{2/3}$) is named after him.

Courtesy of University of Delaware

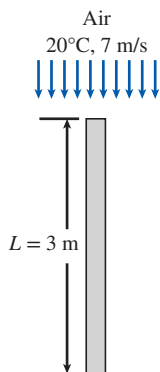


FIGURE 6-42

Schematic for Example 6-4.

for $0.6 < Pr < 60$. Here j_H is called the *Colburn j-factor* (Fig. 6-41). Although this relation is developed using relations for laminar flow over a flat plate (for which $\partial P^*/\partial x^* = 0$), experimental studies show that it is also applicable approximately for turbulent flow over a surface, even in the presence of pressure gradients. For laminar flow, however, the analogy is not applicable unless $\partial P^*/\partial x^* \approx 0$. Therefore, it does not apply to laminar flow in a pipe. Analogies between C_f and Nu that are more accurate are also developed, but they are more complex and beyond the scope of this book. The analogies given above can be used for both local and average quantities.

EXAMPLE 6-4 Finding Convection Coefficient from Drag Measurement

A 2-m $\times$ 3-m flat plate is suspended in a room and is subjected to airflow parallel to its surfaces along its 3-m-long side. The free stream temperature and velocity of air are 20°C and 7 m/s. The total drag force acting on the plate is measured to be 0.86 N. Determine the average convection heat transfer coefficient for the plate (Fig. 6-42).

SOLUTION A flat plate is subjected to airflow, and the drag force acting on it is measured. The average convection coefficient is to be determined.

Assumptions 1 Steady operating conditions exist. 2 The edge effects are negligible. 3 The local atmospheric pressure is 1 atm.

Properties The properties of air at 20°C and 1 atm are (Table A-15):

$$\rho = 1.204 \text{ kg/m}^3, \quad c_p = 1.007 \text{ kJ/kg} \cdot \text{K}, \quad Pr = 0.7309$$

Analysis The flow is along the 3-m side of the plate, and thus the characteristic length is $L = 3 \text{ m}$. Both sides of the plate are exposed to airflow, and thus the total surface area is

$$A_s = 2WL = 2(2 \text{ m})(3 \text{ m}) = 12 \text{ m}^2$$

For flat plates, the drag force is equivalent to friction force. The average friction coefficient C_f can be determined from Eq. 6-11,

$$F_f = C_f A_s \frac{\rho V^2}{2}$$

Solving for C_f and substituting,

$$C_f = \frac{F_f}{\rho A_s V^2 / 2} = \frac{0.86 \text{ N}}{(1.204 \text{ kg/m}^3)(12 \text{ m}^2)(7 \text{ m/s})^2 / 2} \left(\frac{1 \text{ kg} \cdot \text{m/s}^2}{1 \text{ N}} \right) = 0.00243$$

Then the average heat transfer coefficient can be determined from the modified Reynolds analogy (Eq. 6-83) to be

$$h = \frac{C_f \rho V c_p}{2 Pr^{2/3}} = \frac{0.00243 (1.204 \text{ kg/m}^3)(7 \text{ m/s})(1007 \text{ J/kg} \cdot \text{K})}{2 (0.7309)^{2/3}} = 12.7 \text{ W/m}^2 \cdot \text{K}$$

Discussion This example shows the great utility of momentum-heat transfer analogies in that the convection heat transfer coefficient can be obtained from a knowledge of the friction coefficient, which is easier to determine.

TOPIC OF SPECIAL INTEREST

*Microscale Heat Transfer**

Heat transfer considerations play a crucial role in the design and operation of many modern devices. New approaches and methods of analyses have been developed to understand and modulate (enhance or suppress) such energy interactions. Modulation typically occurs through actively controlling the surface phenomena, or focusing of the volumetric energy. In this section we discuss one such example—microscale heat transfer.

Recent inventions in micro ($\sim 10^{-6}$ m) and nano ($\sim 10^{-9}$ m) scale systems have shown tremendous benefits in fluid flow and heat transfer processes. These devices are extremely tiny and are only visible through electron microscopes. The detailed understanding of the governing mechanism of these systems will be at the heart of realizing many future technologies. Examples include chemical and biological sensors, hydrogen storage, space exploration devices, and drug screening. Micro-nanoscale device development also poses several new challenges, however. For example, the classical heat transfer knowledge originates from the thermal equilibrium approach, and the equations are derived for material continua. As the length scale of the system becomes minuscule, the heat transfer through these particles in nanoscale systems is no longer an equilibrium process, and the continuum-based equilibrium approach is no longer valid. Thus, a more general understanding of the concept of heat transfer becomes essential.

Both length and time scales are crucial in micro- and nanoscale heat transfer. The significance of length scale becomes evident from the fact that the surface area per unit volume of an object increases as the length scale of the object shrinks. This means the heat transfer through the surface becomes orders of magnitude more important in microscale than in large everyday objects. Transport of thermal energy in electronic and thermoelectric equipment often occurs at a range of length scales from millimeters to nanometers. For example, in a microelectronic chip (such as the MOSFET in Fig. 6–43) heat is generated in a nanometer-size drain region and ultimately conducted to the surroundings through substrates whose thickness is of the order of a millimeter. Clearly energy transport and conversion mechanisms in devices involve a wide range of length scales and are quite difficult to model.

Small time scales also play an important role in energy transport mechanisms. For example, ultra-short (pico-second and femto-second) pulse lasers are extremely useful for material processing industries. Here the tiny time scales permit localized laser-material interaction beneficial for high-energy deposition and transport.

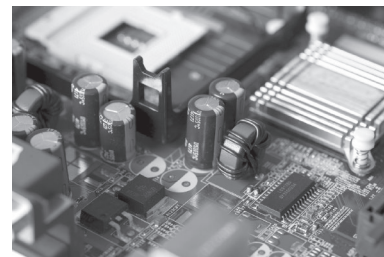
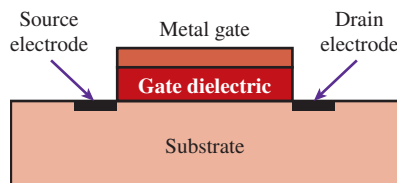
The applicability of the continuum model is determined by the local value of the nondimensional Knudsen number (Kn), which is defined as the ratio of the mean free path (mfp) of the heat-carrier medium to the

*This section is contributed by Subrata Roy, Computational Plasma Dynamics Laboratory, Mechanical Engineering, Kettering University, Flint, MI.

FIGURE 6–43

Metal-oxide semiconductor field-effect transistor (MOSFET) used in microelectronics.

Photo: ©Ferenc Szelepcsényi/Alamy



system reference length scale (say thermal diffusion length). Microscale effects become important when the mfp becomes comparable to or greater than the reference length of the device, say at $Kn > 0.001$. As a result, thermophysical properties of materials become dependent on structure, and heat conduction processes are no longer local phenomena, but rather exhibit long-range radiative effects.

The conventional macroscopic Fourier conduction model violates this nonlocal feature of microscale heat transfer, and alternative approaches are necessary for analysis. The most suitable model to date is the concept of *phonon*. The thermal energy in a uniform solid material can be interpreted as the vibrations of a regular lattice of closely bound atoms inside. These atoms exhibit collective modes of sound waves (phonons) which transport energy at the speed of sound in a material. Following quantum mechanical principles, phonons exhibit particle-like properties of bosons with zero spin (wave-particle duality). Phonons play an important role in many of the physical properties of solids, such as the thermal and the electrical conductivities. In insulating solids, phonons are also the primary mechanism by which heat conduction takes place.

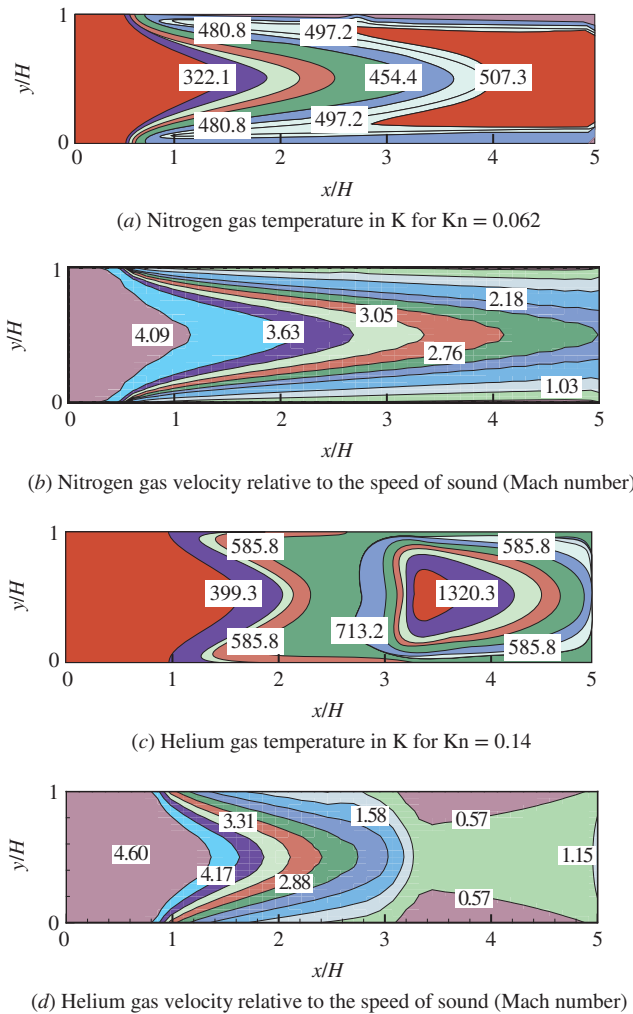
The variation of temperature near the bounding wall continues to be a major determinant of heat transfer though the surface. However, when the continuum approach breaks down, the conventional Newton's law of cooling using wall and bulk fluid temperature needs to be modified. Specifically, unlike in macroscale objects where the wall and adjacent fluid temperatures are equal ($T_w = T_g$), in a micro device there is a temperature slip and the two values are different. One well-known relation for calculating the temperature jump at the wall of a microgeometry was derived by von Smoluchowski in 1898,

$$T_g - T_w = \frac{2 - \sigma_T}{\sigma_T} \left[\frac{2\gamma}{\gamma + 1} \right] \frac{\lambda}{Pr} \left(\frac{\partial T}{\partial y} \right)_w \quad (6-84)$$

where T is the temperature in K, σ_T is the thermal accommodation coefficient and indicates the molecular fraction reflected diffusively from the wall, γ is the specific heat ratio, and Pr is the Prandtl number. Once this value is known, the heat transfer rate may be calculated from:

$$-k \left(\frac{\partial T}{\partial y} \right)_w = \frac{\sigma_T \sqrt{2\pi RT}}{2 - \sigma_T} \left[\frac{\gamma + 1}{2\gamma} \right] \frac{5\rho c_p}{16} (T_w - T_g) \quad (6-85)$$

As an example, the temperature distribution and Mach number contours inside a microtube of width $H = 1.2 \mu\text{m}$ are plotted in Fig. 6–44 for

**FIGURE 6-44**

Fluid-thermal characteristics inside a microchannel.

From Raju and Roy, 2005.

supersonic flow of nitrogen and helium. For nitrogen gas with an inlet $Kn = 0.062$, the gas temperature (T_g) adjacent to the wall differs substantially from the fixed wall temperature, as shown in Fig. 6-44a, where T_w is 323 K and T_g is almost 510 K. The effect of this wall heat transfer is to reduce the Mach number, as shown in Figure 6-44b, but the flow remains supersonic. For helium gas with inlet $Kn = 0.14$ and a lower wall temperature of 298 K, the gas temperature immediately adjacent to the wall is even higher—up to 586 K, as shown in the Fig. 6-44c. This creates very high wall heat flux that is unattainable in macroscale applications. In this case, shown in Fig. 6-44d, heat transfer is large enough to choke the flow.

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SUMMARY

Convection heat transfer is expressed by *Newton's law of cooling* as

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty)$$

where h is the convection heat transfer coefficient, T_s is the surface temperature, and T_∞ is the free-stream temperature. The convection coefficient is also expressed as

$$h = \frac{-k_{\text{fluid}}(\partial T/\partial y)_{y=0}}{T_s - T_\infty}$$

The *Nusselt number*, which is the dimensionless heat transfer coefficient, is defined as

$$\text{Nu} = \frac{hL_c}{k}$$

where k is the thermal conductivity of the fluid and L_c is the characteristic length.

The highly ordered fluid motion characterized by smooth streamlines is called *laminar*. The highly disordered fluid motion that typically occurs at high velocities characterized by velocity fluctuations is called *turbulent*. The random and rapid fluctuations of groups of fluid particles, called *eddies*, provide an additional mechanism for momentum and heat transfer.

The region of the flow above the plate bounded by δ in which the effects of the viscous shearing forces caused by fluid viscosity are felt is called the *velocity boundary layer*. The *boundary layer thickness*, δ , is defined as the distance from the surface at which $u = 0.99V$. The hypothetical line of $u = 0.99V$ divides the flow over a plate into the *boundary layer region* in which the viscous effects and the velocity changes are significant, and the *irrotational flow region*, in which the frictional effects are negligible.

The friction force per unit area is called *shear stress*, and the shear stress at the wall surface is expressed as

$$\tau_w = \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} \quad \text{or} \quad \tau_w = C_f \frac{\rho V^2}{2}$$

where μ is the dynamic viscosity, V is the upstream velocity, and C_f is the dimensionless *friction coefficient*. The property $\nu = \mu/\rho$ is the *kinematic viscosity*. The friction force over the entire surface is determined from

$$F_f = C_f A_s \frac{\rho V^2}{2}$$

The flow region over the surface in which the temperature variation in the direction normal to the surface is significant is the *thermal boundary layer*. The *thickness* of the thermal boundary layer δ_t at any location along the surface is the distance from the surface at which the temperature difference $T - T_s$ equals $0.99(T_\infty - T_s)$. The relative thickness of the velocity and the thermal boundary layers is best described by the dimensionless *Prandtl number*, defined as

$$\text{Pr} = \frac{\text{Molecular diffusivity of momentum}}{\text{Molecular diffusivity of heat}} = \frac{\nu}{\alpha} = \frac{\mu c_p}{k}$$

For external flow, the dimensionless *Reynolds number* is expressed as

$$\text{Re} = \frac{\text{Inertia forces}}{\text{Viscous forces}} = \frac{VL_c}{\nu} = \frac{\rho VL_c}{\mu}$$

For a flat plate, the characteristic length is the distance x from the leading edge. The Reynolds number at which the flow becomes turbulent is called the *critical Reynolds number*. For flow over a flat plate, its value is taken to be $\text{Re}_{\text{cr}} = Vx_{\text{cr}}/\nu = 5 \times 10^5$.

The continuity, momentum, and energy equations for steady two-dimensional incompressible flow with constant properties are determined from mass, momentum, and energy balances to be

$$\text{Continuity:} \quad \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$x\text{-momentum:} \quad \rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = \mu \frac{\partial^2 u}{\partial y^2} - \frac{\partial P}{\partial x}$$

$$\text{Energy:} \quad \rho c_p \left[u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right] = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) + \mu \Phi$$

where the viscous dissipation function Φ is

$$\Phi = 2 \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 \right] + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)^2$$

Using the boundary layer approximations and a similarity variable, these equations can be solved for parallel steady incompressible flow over a flat plate, with the following results:

Velocity boundary layer thickness: $\delta = \frac{4.91}{\sqrt{V/\nu x}} = \frac{4.91x}{\sqrt{\text{Re}_x}}$

Local friction coefficient: $C_{f,x} = \frac{\tau_w}{\rho V^2/2} = 0.664 \text{Re}_x^{-1/2}$

Local Nusselt number: $\text{Nu}_x = \frac{h_x x}{k} = 0.332 \text{Pr}^{1/3} \text{Re}_x^{1/2}$

Thermal boundary layer thickness: $\delta_t = \frac{\delta}{\text{Pr}^{1/3}} = \frac{4.91x}{\text{Pr}^{1/3} \sqrt{\text{Re}_x}}$

The average friction coefficient and Nusselt number are expressed in functional form as

$$C_f = f(\text{Re}_L) \quad \text{and} \quad \text{Nu} = g(\text{Re}_L, \text{Pr})$$

The Nusselt number can be expressed by a simple power-law relation of the form

$$\text{Nu} = C \text{Re}_L^m \text{Pr}^n$$

where m and n are constant exponents, and the value of the constant C depends on geometry. The *Reynolds analogy* relates the convection coefficient to the friction coefficient for fluids with $\text{Pr} \approx 1$ and is expressed as

$$C_{f,x} \frac{\text{Re}_x}{2} = \text{Nu}_x \quad \text{or} \quad \frac{C_{f,x}}{2} = \text{St}_x$$

where

$$\text{St} = \frac{h}{\rho c_p V} = \frac{\text{Nu}}{\text{Re}_L \text{Pr}}$$

is the *Stanton number*. The analogy is extended to other Prandtl numbers by the *modified Reynolds analogy* or *Chilton–Colburn analogy*, expressed as

$$C_{f,x} \frac{\text{Re}_x}{2} = \text{Nu}_x \text{Pr}^{-1/3}$$

or

$$\frac{C_{f,x}}{2} = \text{St}_x \text{Pr}^{2/3} = j_H (0.6 < \text{Pr} < 60)$$

These analogies are also applicable approximately for turbulent flow over a surface, even in the presence of pressure gradients.

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PROBLEMS*

Mechanism and Types of Convection

6-1C Define incompressible flow and incompressible fluid. Must the flow of a compressible fluid necessarily be treated as compressible?

6-2C What is forced convection? How does it differ from natural convection? Is convection that is caused by winds forced or natural convection?

6-3C What is external forced convection? How does it differ from internal forced convection? Can a heat transfer system involve both internal and external convection at the same time? Give an example.

6-4C In which mode of heat transfer is the convection heat transfer coefficient usually higher, natural convection or forced convection? Why?

6-5C Consider a hot baked potato. Will the potato cool faster or slower when we blow the warm air coming from our lungs on it instead of letting it cool naturally in the cooler air in the room? Explain.

6-6C What is the physical significance of the Nusselt number? How is it defined?

6-7C When is heat transfer through a fluid conduction and when is it convection? For what case is the rate of heat transfer higher? How does the convection heat transfer coefficient differ from the thermal conductivity of a fluid?

6-8 An average man has a body surface area of 1.8 m^2 and a skin temperature of 33°C . The convection heat transfer coefficient for a clothed person walking in still air is expressed as $h = 8.6V^{0.53}$ for $0.5 < V < 2 \text{ m/s}$, where V is the walking velocity in m/s . Assuming the average surface temperature of the clothed person to be 30°C , determine the rate of heat loss from an average man walking in still air at 7°C by convection at a walking velocity of (a) 0.5 m/s , (b) 1.0 m/s , (c) 1.5 m/s , and (d) 2.0 m/s .

6-9 The convection heat transfer coefficient for a clothed person standing in moving air is expressed as $h = 14.8V^{0.69}$ for $0.15 < V < 1.5 \text{ m/s}$, where V is the air velocity. For a person with a body surface area of 1.7 m^2 and an average surface temperature of 29°C , determine the rate of heat loss from the person in windy air at 10°C by convection for air velocities of (a) 0.5 m/s , (b) 1.0 m/s , and (c) 1.5 m/s .

*Problems designated by a "C" are concept questions, and students are encouraged to answer them all. Problems designated by an "E" are in English units, and the SI users can ignore them. Problems with the  icon are comprehensive in nature and are intended to be solved with appropriate software. Problems with the  icon are Prevention through Design problems. Problems with the  icon are Codes and Standards problems.

6-10 During air cooling of potatoes, the heat transfer coefficient for combined convection, radiation, and evaporation is determined experimentally to be as shown:

Air Velocity, m/s	Heat Transfer Coefficient, $\text{W/m}^2\cdot\text{K}$
0.66	14.0
1.00	19.1
1.36	20.2
1.73	24.4

Consider an 8-cm -diameter potato initially at 20°C . Potatoes are cooled by refrigerated air at 5°C at a velocity of 1 m/s . Determine the initial rate of heat transfer from a potato, and the initial value of the temperature gradient in the potato at the surface.

6-11 The upper surface of a 50-cm -thick solid plate ($k = 237 \text{ W/m}\cdot\text{K}$) is being cooled by water with temperature of 20°C . The upper and lower surfaces of the solid plate maintained at constant temperatures of 60°C and 120°C , respectively. Determine the water convection heat transfer coefficient and the water temperature gradient at the upper plate surface.

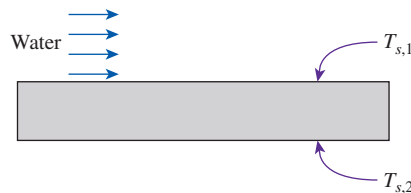


FIGURE P6-11

6-12 Consider airflow over a plate surface maintained at a temperature of 220°C . The temperature profile of the airflow is given as

$$T(y) = T_\infty - (T_\infty - T_s) \exp\left(-\frac{V}{\alpha_{\text{fluid}}} y\right)$$

The airflow at 1 atm has a free stream velocity and temperature of 0.08 m/s and 20°C , respectively. Determine the heat flux on the plate surface and the convection heat transfer coefficient of the airflow.

6-13 During air cooling of oranges, grapefruit, and tangelos, the heat transfer coefficient for combined convection, radiation, and evaporation for air velocities of $0.11 < V < 0.33 \text{ m/s}$ is determined experimentally and is expressed as $h = 5.05 k_{\text{air}} \text{Re}^{1/3}/D$, where the diameter D is the characteristic length. Oranges are cooled by refrigerated air at 3°C and 1 atm at a velocity of 0.3 m/s . Determine (a) the initial rate of heat transfer from a 7-cm -diameter orange initially at 15°C with a thermal

conductivity of $0.7 \text{ W/m}\cdot\text{K}$, (b) the value of the initial temperature gradient inside the orange at the surface, and (c) the value of the Nusselt number.

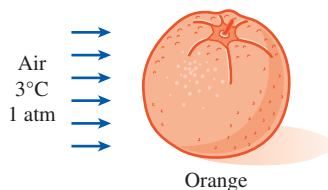


FIGURE P6-13

6-14 During air cooling of steel balls, the convection heat transfer coefficient is determined experimentally as a function of air velocity to be $h = 17.9V^{0.54}$ for $0.5 < V < 2 \text{ m/s}$, where h and V are in $\text{W/m}^2\cdot\text{K}$ and m/s , respectively. Consider a 24-mm-diameter steel ball initially at 300°C with a thermal conductivity of $15 \text{ W/m}\cdot\text{K}$. The steel balls are cooled in air at 10°C and a velocity of 1.5 m/s . Determine the initial values of the heat flux and the temperature gradient in the steel ball at the surface.

6-15  The upper surface of a $1\text{-m} \times 1\text{-m}$ ASTM B152 copper plate is being cooled by air at 20°C . The air is flowing parallel over the plate surface at a velocity of 0.5 m/s . The local convection heat transfer coefficient along the plate surface is correlated with the correlation $h_x = 1.36 x^{-0.5}$, where h_x and x have the units $\text{W/m}^2\cdot\text{K}$ and m , respectively. The maximum use temperature for the ASTM B152 copper plate is 260°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). Evaluate the average convection heat transfer coefficient h for the entire plate. If the rate of convection heat transfer from the plate surface is 700 W , would the use of ASTM B152 plate be in compliance with the ASME Code for Process Piping?

6-16  In cryogenic equipment, cold gas flows in parallel over the surface of a $1\text{-m} \times 1\text{-m}$ ASTM A240 410S stainless steel plate. The average convection heat transfer coefficient for the plate surface is correlated with the gas velocity as $h = 6.5V^{0.8}$, where h and V have the units $\text{W/m}^2\cdot\text{K}$ and m/s , respectively. The temperature of the cold gas is -50°C . The minimum temperature suitable for the ASTM A240 410S plate is -30°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). To keep the plate's temperature from going below -30°C , the plate is heated at a rate of 840 W . Determine the maximum velocity that the gas can achieve without cooling the plate below the suitable temperature set by the ASME Code for Process Piping.

6-17  Two metal plates are connected by a long ASTM B98 copper-silicon bolt. A hot gas at 200°C flows between the plates and across the cylindrical bolt. The diameter of the bolt is 9.5 mm , and the length of the bolt exposed to the hot gas is 10 cm . The average convection heat transfer coefficient for the bolt in crossflow is correlated with the gas

velocity as $h = 24.6V^{0.62}$, where h and V have the units $\text{W/m}^2\cdot\text{K}$ and m/s , respectively. The maximum use temperature for the ASTM B98 bolt is 149°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-2M). If the gas velocity is 10.4 m/s , determine the minimum heat removal rate required to keep the bolt surface from going above the maximum use temperature.

6-18  Two metal plates are connected by a long ASTM A479 904L stainless steel bar. A hot gas, at 400°C , flows between the plates and across the bar. The bar has a square cross section with a width of 2 cm , and the length of the bar exposed to the hot gas is 10 cm . The average convection heat transfer coefficient for the bar in crossflow is correlated with the gas velocity as $h = 13.6V^{0.675}$, where h and V have the units $\text{W/m}^2\cdot\text{K}$ and m/s , respectively. The maximum use temperature for the ASTM A479 904L is 260°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). The temperature of the bar is maintained by a cooling mechanism with the capability of removing heat at a rate of 100 W . Determine the maximum velocity that the gas can achieve without heating the stainless steel bar above the maximum use temperature set by the ASME Code for Process Piping.

6-19  A ball bearing manufacturing plant is using air to cool chromium steel balls ($k = 40 \text{ W/m}\cdot\text{K}$). The convection heat transfer coefficient for the cooling is determined experimentally as a function of air velocity to be $h = 18.05V^{0.56}$, where h and V are in $\text{W/m}^2\cdot\text{K}$ and m/s , respectively. At a given moment during the cooling process with the air temperature at 5°C , a chromium steel ball has a surface temperature of 450°C . Using appropriate software, determine the effect of the air velocity (V) on the temperature gradient in the chromium steel ball at the surface. By varying the air velocity from 0.2 to 2.4 m/s with increments of 0.2 m/s , plot the temperature gradient in the chromium steel ball at the surface as a function of air velocity.

6-20 Consider the surface of a metal plate being cooled by forced convection. Determine the ratio of the temperature gradient in the fluid to the temperature gradient in the metal plate at the surface.

6-21 A metal plate is being cooled by air ($k_{\text{fluid}} = 0.259 \text{ W/m}\cdot\text{K}$) at the upper surface while the lower surface is subjected to a uniform heat flux of 1000 W/m^2 . Determine the temperature gradient in the air at the upper surface of the metal plate.

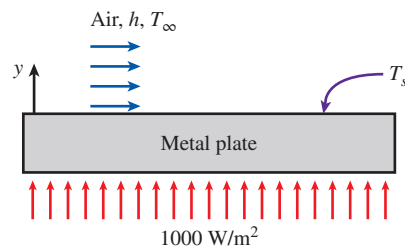


FIGURE P6-21

6-22 The top surface of a metal plate ($k_{\text{plate}} = 237 \text{ W/m}\cdot\text{K}$) is being cooled by air ($k_{\text{air}} = 0.243 \text{ W/m}\cdot\text{K}$) while the bottom surface is exposed to a hot steam at 100°C with a convection heat transfer coefficient of $30 \text{ W/m}^2\cdot\text{K}$. If the bottom surface temperature of the plate is 80°C , determine the temperature gradient in the air and the temperature gradient in the plate at the top surface of the plate.

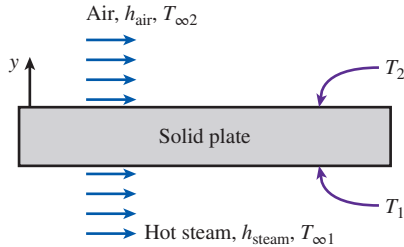


FIGURE P6-22

6-23 During air cooling of a flat plate ($k = 1.4 \text{ W/m}\cdot\text{K}$), the convection heat transfer coefficient is given as a function of air velocity to be $h = 27V^{0.85}$, where h and V are in $\text{W/m}^2\cdot\text{K}$ and m/s , respectively. At a given moment, the surface temperature of the plate is 75°C and the air ($k = 0.0266 \text{ W/m}\cdot\text{K}$) temperature is 5°C . Using appropriate software, determine the effect of the air velocity (V) on the air temperature gradient at the plate surface. By varying the air velocity from 0 to 1.2 m/s with increments of 0.1 m/s , plot the air temperature and plate temperature gradients at the plate surface as a function of air velocity.

6-24 A metal plate ($k = 180 \text{ W/m}\cdot\text{K}$, $\rho = 2800 \text{ kg/m}^3$, and $c_p = 880 \text{ J/kg}\cdot\text{K}$) with a thickness of 1 cm is being cooled by air at 5°C with a convection heat transfer coefficient of $30 \text{ W/m}^2\cdot\text{K}$. If the initial temperature of the plate is 300°C , determine the plate temperature gradient at the surface after 2 minutes of cooling. *Hint:* Use the lumped system analysis to calculate the plate surface temperature. Make sure to verify the application of this method to this problem.

6-25 Air at 5°C , with a convection heat transfer coefficient of $30 \text{ W/m}^2\cdot\text{K}$, is used for cooling metal plates coming out of a heat treatment oven at an initial temperature of 300°C . The plates ($k = 180 \text{ W/m}\cdot\text{K}$, $\rho = 2800 \text{ kg/m}^3$, and $c_p = 880 \text{ J/kg}\cdot\text{K}$) have a thickness of 10 mm. Using appropriate software, determine the effect of cooling time on the temperature gradient in the metal plates at the surface. By varying the cooling time from 0 to 3000 s, plot the temperature gradient in the plates at the surface as a function of cooling time. *Hint:* Use the lumped system analysis to calculate the plate surface temperature. Make sure to verify the application of this method to this problem.

6-26 A 5-mm-thick stainless steel strip ($k = 21 \text{ W/m}\cdot\text{K}$, $\rho = 8000 \text{ kg/m}^3$, and $c_p = 570 \text{ J/kg}\cdot\text{K}$) is being heat treated as it moves through a furnace at a speed of 1 cm/s . The air

temperature in the furnace is maintained at 900°C with a convection heat transfer coefficient of $80 \text{ W/m}^2\cdot\text{K}$. If the furnace length is 3 m and the stainless steel strip enters it at 20°C , determine the surface temperature gradient of the strip at mid-length of the furnace. *Hint:* Use the lumped system analysis to calculate the plate surface temperature. Make sure to verify the application of this method to this problem.

6-27 A long steel strip is being conveyed through a 3-m-long furnace to be heat treated at a speed of 0.01 m/s . The steel strip ($k = 21 \text{ W/m}\cdot\text{K}$, $\rho = 8000 \text{ kg/m}^3$, and $c_p = 570 \text{ J/kg}\cdot\text{K}$) has a thickness of 5 mm, and it enters the furnace at an initial temperature of 20°C . Inside the furnace, the air temperature is maintained at 900°C with a convection heat transfer coefficient of $80 \text{ W/m}^2\cdot\text{K}$. Using appropriate software, determine the surface temperature gradient of the steel strip as a function of location inside the furnace. By varying the location in the furnace for $0 \leq x \leq 3 \text{ m}$ with increments of 0.2 m, plot the surface temperature gradient of the strip as a function of furnace location. *Hint:* Use the lumped system analysis to calculate the plate surface temperature. Make sure to verify the application of this method to this problem.

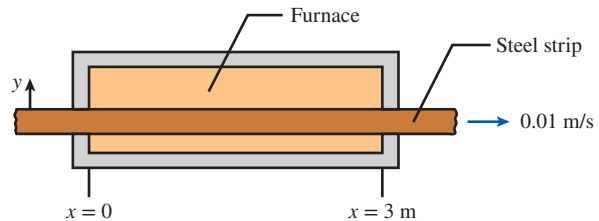


FIGURE P6-27

Boundary Layers and Flow Regimes

6-28C What is the no-slip condition? What causes it?

6-29C What is Newtonian fluid? Is water a Newtonian fluid?

6-30C What is viscosity? What causes viscosity in liquids and in gases? Is dynamic viscosity typically higher for a liquid or for a gas?

6-31C Consider two identical small glass balls dropped into two identical containers, one filled with water and the other with oil. Which ball will reach the bottom of the container first? Why?

6-32C How does the dynamic viscosity of (a) liquids and (b) gases vary with temperature?

6-33C What fluid property is responsible for the development of the velocity boundary layer? For what kinds of fluids will there be no velocity boundary layer on a flat plate?

6-34C What is the physical significance of the Prandtl number? Does the value of the Prandtl number depend on the type of flow or the flow geometry? Does the Prandtl number of air change with pressure? Does it change with temperature?

6-35C Will a thermal boundary layer develop in flow over a surface even if both the fluid and the surface are at the same temperature?

6-36C What is the physical significance of the Reynolds number? How is it defined for external flow over a plate of length L ?

6-37C How does turbulent flow differ from laminar flow? For which flow is the heat transfer coefficient higher?

6-38C What does the friction coefficient represent in flow over a flat plate? How is it related to the drag force acting on the plate?

6-39C What is the physical mechanism that causes the friction factor to be higher in turbulent flow?

6-40C What is turbulent viscosity? What is it caused by?

6-41C What is turbulent thermal conductivity? What is it caused by?

6-42 Consider fluid flow over a surface with a velocity profile given as

$$u(y) = 100(y + 2y^2 - 0.5y^3) \text{ m/s}$$

Determine the shear stress at the wall surface, if the fluid is (a) air at 1 atm and (b) liquid water, both at 20°C. Also calculate the wall shear stress ratio for the two fluids, and interpret the result.

6-43 Consider a flow over a surface with the velocity and temperature profiles given as

$$\begin{aligned} u(y) &= C_1(y + y^2 - y^3) \\ T(y) &= C_2 - e^{-2C_2y} \end{aligned}$$

where the coefficients C_1 and C_2 are constants. Determine the expressions for the friction coefficient (C_f) and the convection heat transfer coefficient (h).

6-44 Air flowing over a 1-m-long flat plate at a velocity of 7 m/s has a friction coefficient given as $C_f = 0.664(Vx/\nu)^{-0.5}$, where x is the location along the plate. Determine the wall shear stress and the air velocity gradient on the plate surface at mid-length of the plate. Evaluate the air properties at 20°C and 1 atm.

6-45  The friction coefficient of air flowing over a flat plate is given as $C_f = 0.664(Vx/\nu)^{-0.5}$, where x is the location along the plate. Using appropriate software, determine the effect of the air velocity (V) on the wall shear stress (τ_w) at the plate locations of $x = 0.5$ m and 1 m. By varying the air velocity from 0.5 to 6 m/s with increments of 0.5 m/s, plot the wall shear stress as a function of air velocity at $x = 0.5$ m and 1 m. Evaluate the air properties at 20°C and 1 atm.

6-46  Air flowing over a flat plate at 5 m/s has a friction coefficient given as $C_f = 0.664(Vx/\nu)^{-0.5}$, where x is the location along the plate. Using appropriate software, determine the effect of the location along the plate (x) on

the wall shear stress (τ_w). By varying x from 0.01 to 1 m, plot the wall shear stress as a function of x . Evaluate the air properties at 20°C and 1 atm.

6-47 Consider a flat plate positioned inside a wind tunnel, and air at 1 atm and 20°C is flowing with a free stream velocity of 60 m/s. What is the minimum length of the plate necessary for the Reynolds number to reach 2×10^7 ? If the critical Reynolds number is 5×10^5 , what type of flow regime would the airflow experience at 0.2 m from the leading edge?

6-48 Air flows over a flat plate at 40 m/s, 25°C, and 1 atm pressure. (a) What plate length should be used to achieve a Reynolds number of 1×10^8 at the end of the plate? (b) If the critical Reynolds number is 5×10^5 , at what distance from the leading edge of the plate would transition occur?

6-49 Consider fluid flowing with a free stream velocity of 5 m/s over a flat plate, where the critical Reynolds number is 5×10^5 . Determine the distance from the leading edge at which the transition from laminar to turbulent flow occurs for air (at 1 atm), liquid water, methanol, and engine oil, all at 20°C, and mercury at 25°C.

6-50E Consider fluid flowing with a free stream velocity of 1 ft/s over a flat plate, where the critical Reynolds number is 5×10^5 . Determine the distance from the leading edge at which the transition from laminar to turbulent flow occurs for air (at 1 atm), liquid water, isobutane, and engine oil, and mercury. Evaluate all properties at 50°F.

Convection Equations and Similarity Solutions

6-51C Consider steady, laminar, two-dimensional, incompressible flow with constant properties and a Prandtl number of unity. For a given geometry, is it correct to say that both the average friction and heat transfer coefficients depend on the Reynolds number only?

6-52C Express the continuity equation for steady two-dimensional flow with constant properties, and explain what each term represents.

6-53C Is the acceleration of a fluid particle necessarily zero in steady flow? Explain.

6-54C For steady two-dimensional flow, what are the boundary layer approximations?

6-55C For what types of fluids and flows is the viscous dissipation term in the energy equation likely to be significant?

6-56C For steady two-dimensional flow over an isothermal flat plate in the x -direction, express the boundary conditions for the velocity components u and v and the temperature T at the plate surface and at the edge of the boundary layer.

6-57C What is a similarity variable, and what is it used for? For what kinds of functions can we expect a similarity solution for a set of partial differential equations to exist?

6-58C Consider steady, laminar, two-dimensional flow over an isothermal plate. Does the thickness of the velocity boundary layer increase or decrease with (a) distance from the leading edge, (b) free-stream velocity, and (c) kinematic viscosity?

6-59C Consider steady, laminar, two-dimensional flow over an isothermal plate. Does the wall shear stress increase, decrease, or remain constant with distance from the leading edge?

6-60C What are the advantages of nondimensionalizing the convection equations?

6-61C Under what conditions can a curved surface be treated as a flat plate in fluid flow and convection analysis?

6-62 Consider a 5-cm-diameter shaft rotating at 5600 rpm in a 25-cm-long bearing with a clearance of 0.5 mm. Determine the power required to rotate the shaft if the fluid in the gap is (a) air, (b) water, and (c) oil at 40°C and 1 atm.

6-63 Oil flow in a journal bearing can be treated as parallel flow between two large isothermal plates with one plate moving at a constant velocity of 8 m/s and the other stationary. Consider such a flow with a uniform spacing of 0.7 mm between the plates. The temperatures of the upper and lower plates are 40°C and 15°C, respectively. By simplifying and solving the continuity, momentum, and energy equations, determine (a) the velocity and temperature distributions in the oil, (b) the maximum temperature and where it occurs, and (c) the heat flux from the oil to each plate.

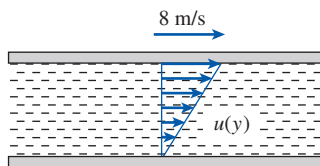


FIGURE P6-63

6-64 Repeat Prob. 6-63 for a spacing of 0.5 mm.

6-65 Consider the flow of fluid between two large parallel isothermal plates separated by a distance L . The upper plate is moving at a constant velocity of V and maintained at temperature T_0 while the lower plate is stationary and insulated. By simplifying and solving the continuity, momentum, and energy equations, obtain relations for the maximum temperature of fluid, the location where it occurs, and heat flux at the upper plate.

6-66 Reconsider Prob. 6-65. Using the results of this problem, obtain a relation for the volumetric heat generation rate $\dot{e}_{\text{gen}}$, in W/m^3 . Then express the convection problem as an equivalent conduction problem in the oil layer. Verify your model by solving the conduction problem and obtaining a relation for the maximum temperature, which should be identical to the one obtained in the convection analysis.

6-67 A 6-cm-diameter shaft rotates at 3000 rpm in a 20-cm-long bearing with a uniform clearance of 0.2 mm. At

steady operating conditions, both the bearing and the shaft in the vicinity of the oil gap are at 50°C, and the viscosity and thermal conductivity of lubricating oil are $0.05 \text{ N}\cdot\text{s/m}^2$ and $0.17 \text{ W/m}\cdot\text{K}$. By simplifying and solving the continuity, momentum, and energy equations, determine (a) the maximum temperature of oil, (b) the rates of heat transfer to the bearing and the shaft, and (c) the mechanical power wasted by the viscous dissipation in the oil. Treat this problem as parallel flow between two large plates with one plate moving at constant velocity and the other stationary. **Answers: (a) 53.3°C, (b) 419 W, (c) 838 W**

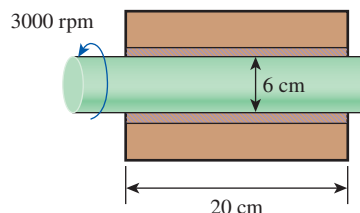


FIGURE P6-67

6-68 Repeat Prob. 6-67 by assuming the shaft to have reached peak temperature and thus heat transfer to the shaft to be negligible, and the bearing surface still to be maintained at 50°C.

6-69  Reconsider Prob. 6-67. Using appropriate software, investigate the effect of shaft velocity on the mechanical power wasted by viscous dissipation. Let the shaft rotation vary from 0 rpm to 6000 rpm. Plot the power wasted versus the shaft rpm, and discuss the results.

6-70 A 4-cm-diameter shaft rotates at 5200 rpm in a 15-cm-long, 8-cm-outer-diameter cast iron bearing ($k = 70 \text{ W/m}\cdot\text{K}$) with a uniform clearance of 0.6 mm filled with lubricating oil ($\mu = 0.03 \text{ N}\cdot\text{s/m}^2$ and $k = 0.14 \text{ W/m}\cdot\text{K}$). The bearing is cooled externally by a liquid, and its outer surface is maintained at 40°C. Disregarding heat conduction through the shaft and assuming one-dimensional heat transfer, determine (a) the rate of heat transfer to the coolant, (b) the surface temperature of the shaft, and (c) the mechanical power wasted by the viscous dissipation in oil. Treat this problem as parallel flow between two large plates with one plate moving at constant velocity and the other stationary.

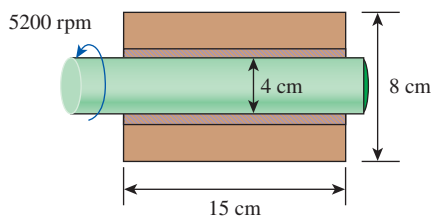


FIGURE P6-70

6-71 Repeat Prob. 6-70 for a clearance of 1 mm.

6-72E Glycerin at 50°F is flowing over a flat plate at a free stream velocity of 6 ft/s. Determine the velocity and thermal boundary layer thicknesses at a distance of 0.5 ft from the leading edge. Also calculate the ratio of the velocity boundary thickness to the thermal boundary layer thickness for this flow, and interpret the result.

6-73 Water at 20°C is flowing with velocity of 0.5 m/s between two parallel flat plates placed 1 cm apart. Determine the distances from the entrance at which the velocity and thermal boundary layers meet.

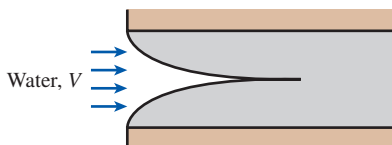


FIGURE P6-73

6-74E Consider a laminar boundary layer flow over a flat plate. Determine the δ/δ_t ratios for air (at 1 atm), liquid water, isobutane, engine oil, and mercury. Evaluate all properties at 50°F.

6-75 For laminar boundary layers, it is reasonable to expect that $\delta/\delta_t \approx \text{Pr}^n$, where n is a positive exponent. Consider laminar boundary layer flow over a flat plate with air at 100°C and 1 atm, where the thermal boundary layer thickness is approximately 15% larger than the velocity boundary layer thickness. Determine the ratio of δ/δ_t if the fluid is engine oil (unused) under the same flow conditions.

6-76 Air at 10°C and 1 atm is flowing over a 0.3-m-long plate at 70°C at velocity of 3.0 m/s. Using appropriate software, plot the following on a combined graph for the range of $x = 0.0$ m to $x = x_{cr}$.

- The hydrodynamic boundary layer as a function of x .
- The thermal boundary layer as a function of x .

6-77 Liquid water at 15°C is flowing over a 0.3-m-wide plate at 65°C at a velocity of 3.0 m/s. Using appropriate software, plot (a) the hydrodynamic boundary layer and (b) the thermal boundary layer as a function of x on the same graph for the range of $x = 0.0$ m to $x = x_{cr}$. Use a critical Reynolds number of 500,000.

6-78 Saturated liquid water at 5°C is flowing over a flat plate at a velocity of 1 m/s. Using appropriate software, determine the effect of the location along the plate (x) on the velocity and thermal boundary layer thicknesses. By varying x for $0 < x \leq 0.5$ m, plot the velocity and thermal boundary layer thicknesses as a function of x . Discuss the results.

6-79 Mercury at 0°C is flowing over a flat plate at a velocity of 0.1 m/s. Using appropriate software, determine the effect of the location along the plate (x) on the

velocity and thermal boundary layer thicknesses. By varying x for $0 < x \leq 0.5$ m, plot the velocity and thermal boundary layer thicknesses as a function of x . Discuss the results.

6-80 Water vapor at 0°C and 1 atm is flowing over a flat plate at a velocity of 10 m/s. Using appropriate software, determine the effect of the location along the plate (x) on the velocity and thermal boundary layer thicknesses. By varying x for $0 < x \leq 0.5$ m, plot the velocity and thermal boundary layer thicknesses as a function of x . Discuss the results.

6-81 Consider a laminar ideal gas flow over a flat plate, where the local Nusselt number can be expressed as $\text{Nu}_x = 0.332 \text{Re}_x^{1/2} \text{Pr}^{1/3}$. Using the expression for the local Nusselt number, show that it can be rewritten in terms of local convection heat transfer coefficient as $h_x = C[V/(xT)]^m$, where C and m are constants.

6-82 Consider air flowing over a 1-m-long flat plate at a velocity of 3 m/s. Determine the convection heat transfer coefficients and the Nusselt numbers at $x = 0.5$ m and 0.75 m. Evaluate the air properties at 40°C and 1 atm.

6-83 A cryogenic gas flows at 5 m/s in parallel over the surface of an ASTM A240 410S stainless steel plate. The temperature of the cold gas is -50°C, and the plate length is 1 m. The minimum temperature suitable for the ASTM A240 410S plate is -30°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). In the interest of designing a heater, determine the total heat flux on the plate surface necessary to maintain the surface temperature at -30°C. Use the following gas properties for the analysis: $c_p = 1.002$ kJ/kg·K, $k = 0.02057$ W/m·K, $\mu = 1.527 \times 10^{-5}$ kg/m·s, and $\rho = 1.514$ kg/m³.

6-84 The upper surface of an ASME SB-96 copper-silicon plate is subjected to convection with hot air flowing at 7.5 m/s parallel over the plate surface. The length of the plate is 1 m, and the temperature of the hot air is 200°C. The ASME Boiler and Pressure Vessel Code (ASME BPVC.IV-2015, HF-300) limits equipment constructed with ASME SB-96 plate to be operated at a temperature not exceeding 93°C. In the interest of designing a cooling mechanism to keep the plate surface temperature from exceeding 93°C, determine the variation of the local heat flux on the plate surface for $0 < x \leq 1$ m. Plot the local heat flux on the plate surface for $0 < x \leq 1$ m. Use the following air properties for the analysis: $c_p = 1.016$ kJ/kg·K, $k = 0.03419$ W/m·K, $\mu = 2.371 \times 10^{-5}$ kg/m·s, and $\rho = 0.8412$ kg/m³.

6-85 The upper surface of an ASTM B152 copper plate is subjected to convection with hot air flowing at 6 m/s parallel over the plate surface. The length of the plate is 1 m, and the temperature of the hot air is 400°C. The maximum use temperature for the ASTM B152 copper plate is 260°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). Determine the variation of the local heat flux with the

thermal boundary layer thickness on the plate surface for $0 < x \leq 1$ m:

- Plot the thermal boundary layer thickness on the plate surface for $0 < x \leq 1$ m.
- Plot the local heat flux as a function of the thermal boundary layer thickness on the plate surface.

Discuss the results. Use the following air properties for the analysis: $c_p = 1.052$ kJ/kg·K, $k = 0.04601$ W/m·K, $\mu = 3.035 \times 10^{-5}$ kg/m·s, and $\rho = 0.5852$ kg/m³.

6-86  Air with a temperature of 20°C is flowing over a flat plate ($k = 15$ W/m·K) at a velocity of 3 m/s. The plate surface temperature is maintained at 60°C. Using appropriate software, determine the effect of the location along the plate (x) on the heat transfer coefficient and the surface temperature gradient of the plate. By varying x for $0 < x \leq 0.5$ m, plot the heat transfer coefficient and the surface temperature gradient of the plate as a function of x . Evaluate the air properties at 40°C and 1 atm.

6-87 For laminar flow over a flat plate, the local heat transfer coefficient varies as $h_x = Cx^{-0.5}$, where x is measured from the leading edge of the plate and C is a constant. Determine the ratio of the average convection heat transfer coefficient over the entire plate of length L to the local convection heat transfer coefficient at the end of the plate.

6-88E An airfoil with a characteristic length of 0.2 ft is placed in airflow at 1 atm and 60°F with free stream velocity of 150 ft/s and convection heat transfer coefficient of 21 Btu/h·ft²·°F. If a second airfoil with a characteristic length of 0.4 ft is placed in the airflow at 1 atm and 60°F with free stream velocity of 75 ft/s, determine the heat flux from the second airfoil. Both airfoils are maintained at a constant surface temperature of 180°F.

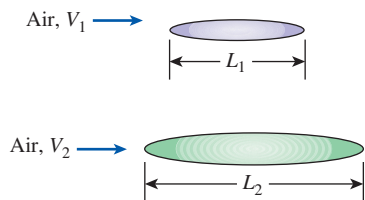


FIGURE P6-88E

6-89 Consider two irregularly shaped objects with different characteristic lengths. The characteristic length of the first object is $L_1 = 0.5$ m, and it is maintained at a uniform surface temperature of $T_{s,1} = 350$ K. The first object is placed in atmospheric air at a temperature of $T_{\infty,1} = 250$ K and an air velocity of $V_1 = 20$ m/s. The average heat flux from the first object under these conditions is 8000 W/m². The second object has a characteristic length of $L_2 = 2.5$ m, is maintained at a uniform surface temperature of $T_{s,2} = 350$ K, and is placed in atmospheric air at a temperature of $T_{\infty,2} = 250$ K and an air velocity of $V_2 = 4$ m/s. Determine the average convection heat transfer coefficient for the second object.

Momentum and Heat Transfer Analogies

6-90C How is the Reynolds analogy expressed? What is the value of it? What are its limitations?

6-91C How is the modified Reynolds analogy expressed? What is the value of it? What are its limitations?

6-92 Consider an airplane cruising at an altitude of 10 km where standard atmospheric conditions are -50°C and 26.5 kPa at a speed of 800 km/h. Each wing of the airplane can be modeled as a 25-m $\times$ 3-m flat plate, and the friction coefficient of the wings is 0.0016. Using the momentum–heat transfer analogy, determine the heat transfer coefficient for the wings at cruising conditions. *Answer: 89.6 W/m²·K*

6-93 A metallic airfoil of elliptical cross section has a mass of 50 kg, surface area of 12m², and a specific heat of 0.50 kJ/kg·K. The airfoil is subjected to airflow at 1 atm, 25°C, and 5 m/s along its 3-m-long side. The average temperature of the airfoil is observed to drop from 160°C to 150°C within 2 min of cooling. Assuming the surface temperature of the airfoil to be equal to its average temperature and using the momentum–heat transfer analogy, determine the average friction coefficient of the airfoil surface. Evaluate the air properties at 25°C and 1 atm. *Answer: 0.000363*

6-94 Repeat Prob. 6-93 for an airflow velocity of 10 m/s.

6-95 The electrically heated 0.6-m-high and 1.8-m-long windshield of a car is subjected to parallel winds at 1 atm, 0°C, and 80 km/h. The electric power consumption is observed to be 70 W when the exposed surface temperature of the windshield is 4°C. Disregarding radiation and heat transfer from the inner surface and using the momentum–heat transfer analogy, determine drag force the wind exerts on the windshield. Evaluate the air properties at 2°C and 1 atm.

6-96 A 5-m $\times$ 5-m flat plate maintained at a constant temperature of 80°C is subjected to parallel flow of air at 1 atm, 20°C, and 10 m/s. The total drag force acting on the upper surface of the plate is measured to be 2.4 N. Using the momentum–heat transfer analogy, determine the average convection heat transfer coefficient and the rate of heat transfer between the upper surface of the plate and the air. Evaluate the air properties at 50°C and 1 atm.

6-97 Air (1 atm, 5°C) with free stream velocity of 2 m/s is flowing in parallel to a stationary thin 1-m $\times$ 1-m flat plate over the top and bottom surfaces. The flat plate has a uniform surface temperature of 35°C. If the friction force asserted on the flat plate is 0.1 N, determine the rate of heat transfer from the plate. Evaluate the air properties at 20°C and 1 atm. *Answer: 1862 W*



FIGURE P6-97

6-98 Air at 1 atm and 20°C is flowing over the top surface of a 0.2-m × 0.5-m-thin metal foil. The airstream velocity is 100 m/s and the metal foil is heated electrically with a uniform heat flux of 6100 W/m². If the friction force on the metal foil surface is 0.3 N, determine the surface temperature of the metal foil. Evaluate the air properties at 100°C and 1 atm. Is this a good assumption?

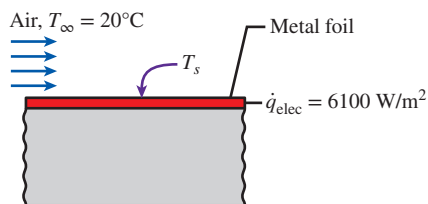


FIGURE P6-98

6-99  The upper surface of an ASME SB-96 copper-silicon plate is subjected to convection with hot air flowing at 6.5 m/s parallel over the plate surface. The plate is square with a length of 1 m, and the temperature of the hot air is 200°C. The ASME Boiler and Pressure Vessel Code (ASME BPVC.IV-2015, HF-300) limits equipment constructed with ASME SB-96 plate to be operated at a temperature not exceeding 93°C. From a wind tunnel experiment, the average friction coefficient for the upper surface of the plate was found to be 0.0023. In the interest of designing a cooling mechanism to keep the plate surface temperature from exceeding 93°C, determine the minimum heat removal rate required to keep the plate surface from going above 93°C. Use the following air properties for the analysis: $c_p = 1.016$ kJ/kg·K, $k = 0.03419$ W/m·K, $\mu = 2.371 \times 10^{-5}$ kg/m·s, and $\rho = 0.8412$ kg/m³.

6-100 Air at 1 atm is flowing over a flat plate with a free stream velocity of 70 m/s. If the convection heat transfer coefficient can be correlated by $Nu_x = 0.03 Re_x^{0.8} Pr^{1/3}$, determine the friction coefficient and wall shear stress at a location 2 m from the leading edge. Evaluate air properties at 20°C and 1 atm.

6-101 Metal plates are being cooled with air blowing in parallel over each plate. The average friction coefficient over each plate is given as $C_f = 1.33 (Re_L)^{-0.5}$ for $Re_L < 5 \times 10^5$. Each metal plate length parallel to the airflow is 1 m. Determine the average convection heat transfer coefficient for the plate if the air velocity is 5 m/s. Evaluate the air properties at 20°C and 1 atm.

6-102  A flat plate is subject to airflow parallel to its surface. The average friction coefficient over the plate is given as

$$C_f = 1.33 (Re_L)^{-1/2} \quad \text{for } Re_L < 5 \times 10^5 \text{ (laminar flow)}$$

$$C_f = 0.074 (Re_L)^{-1/5} \quad \text{for } 5 \times 10^5 \leq Re_L \leq 10^7 \text{ (turbulent flow)}$$

The plate length parallel to the airflow is 1 m. Using appropriate software, determine the effect of air velocity on the average convection heat transfer coefficient for the plate. By varying the air velocity for $0 < V \leq 20$ m/s, plot the average convection heat transfer coefficient as a function of air velocity. Evaluate the air properties at 20°C and 1 atm.

Special Topic: Microscale Heat Transfer

6-103 Using a cylinder, a sphere, and a cube as examples, show that the rate of heat transfer is inversely proportional to the nominal size of the object. That is, heat transfer per unit area increases as the size of the object decreases.

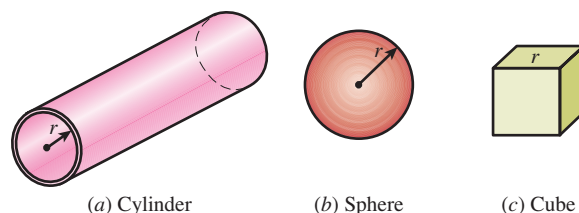


FIGURE P6-103

6-104 Determine the heat flux at the wall of a microchannel of width 1 μm if the wall temperature is 50°C and the average gas temperature near the wall is 100°C for the cases of

- (a) $\sigma_T = 1.0$, $\gamma = 1.667$, $k = 0.15$ W/m·K, $\lambda/Pr = 0.5$
- (b) $\sigma_T = 0.8$, $\gamma = 2$, $k = 0.1$ W/m·K, $\lambda/Pr = 5$

6-105 If $(\partial T/\partial y)_w = 65$ K/m, calculate the Nusselt number for a microchannel of width 1.2 μm if the wall temperature is 50°C and it is surrounded by (a) ambient air at temperature 30°C, (b) nitrogen gas at temperature -100°C.

Review Problems

6-106E Evaluate the Prandtl number from the following data: $c_p = 0.5$ Btu/lbm·R, $k = 2$ Btu/h·ft·R, $\mu = 0.3$ lbm/ft·s.

6-107 A fluid flows at 5 m/s over a wide flat plate 15 cm long. For each substance from the following list, calculate the Reynolds number at the downstream end of the plate. Indicate whether the flow at that point is laminar or turbulent. Assume all fluids are at 50°C. (a) Air, (b) CO₂, (c) Water, (d) Engine oil (unused).

6-108E Consider a fluid flowing over a flat plate at a constant free stream velocity. The critical Reynolds number is 5×10^5 , and the distance from the leading edge at which the transition from laminar to turbulent flow occurs is $x_{cr} = 7$ ft. Determine the characteristic length (L_c) at which the Reynolds number is 1×10^5 .

6-109 Consider the Couette flow of a fluid with a viscosity of $\mu = 0.8$ N·s/m² and thermal conductivity of $k_f = 0.145$ W/m·K. The lower plate is stationary and made of a material of thermal conductivity $k_p = 1.5$ W/m·K and thickness $b = 3$ mm. Its outer surface is maintained at $T_s = 50^\circ\text{C}$. The upper plate

is insulated and moves with a uniform speed $V = 7$ m/s. The distance between plates is $L = 5$ mm.

- Determine the temperature distribution function, $T(y)$, in the fluid ($0 < y < L$).
- Calculate the maximum temperature of the fluid, as well as the temperature of the fluid at the contact surfaces with the lower and upper plates.

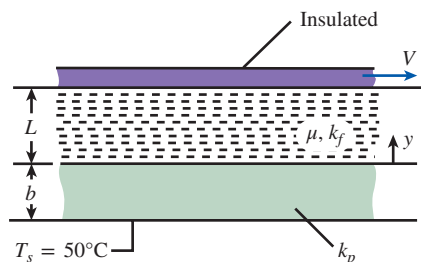


FIGURE P6-109

6-110  Engine oil at 15°C is flowing over a 0.3-m -wide plate at 65°C at a velocity of 3.0 m/s. Using appropriate software, plot (a) the hydrodynamic boundary layer and (b) the thermal boundary layer as a function of x on the same graph for the range of $x = 0.0$ m to $x = x_{cr}$. Use a critical Reynolds number of $500,000$.

6-111 Object 1 with a characteristic length of 0.5 m is placed in airflow at 1 atm and 20°C with free stream velocity of 50 m/s. The heat flux transfer from object 1 when placed in the airflow is measured to be $12,000$ W/m 2 . If object 2 has the same shape and geometry as object 1 (but with a characteristic length of 5 m) and it is placed in the airflow at 1 atm and 20°C with free stream velocity of 5 m/s, determine the average convection heat transfer coefficient for object 2. Both objects are maintained at a constant surface temperature of 120°C .

6-112 A rectangular bar with a characteristic length of 0.5 m is placed in a free stream flow where the convection heat transfer coefficients were found to be 100 W/m $^2\cdot\text{K}$ and 50 W/m $^2\cdot\text{K}$ when the free stream velocities were 25 m/s and 5 m/s, respectively. If the Nusselt number can be expressed as $\text{Nu} = C \text{Re}^m \text{Pr}^n$, where C , m , and n are constants, determine the convection heat transfer coefficients for similar bars with (a) $L = 1$ m and $V = 5$ m/s, and (b) $L = 2$ m and $V = 50$ m/s.

6-113 In an effort to prevent the formation of ice on the surface of a wing, electrical heaters are embedded inside the wing. With a characteristic length of 2.5 m, the wing has a friction coefficient of 0.001 . If the wing is moving at a speed of 200 m/s through air at 1 atm and -20°C , determine the heat flux necessary to keep the wing surface above 0°C . Evaluate the air properties at -10°C and 1 atm.

6-114 A $15\text{-cm} \times 20\text{-cm}$ circuit board is being cooled by forced convection of air at 1 atm. The heat from the circuit board is estimated to be 1000 W/m 2 . If the airstream velocity

is 3 m/s and the shear stress of the circuit board surface is 0.075 N/m 2 , determine the temperature difference between the circuit board surface temperature and the airstream temperature. Evaluate the air properties at 40°C and 1 atm.

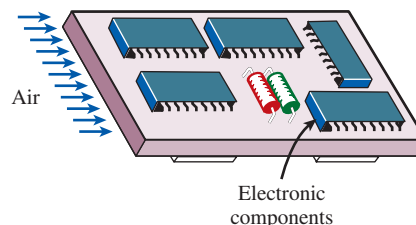


FIGURE P6-114

6-115 Atmospheric air with a velocity of 10 m/s and a temperature of 20°C is in parallel flow over a flat heater surface which is maintained at 80°C . The surface area of the heater is 0.30 m 2 . The drag force induced by the airflow on the heater is measured to be 0.2 N. Using the momentum–heat transfer analogy, determine the electrical power needed to maintain the prescribed heater surface temperature of 80°C . Evaluate the air properties at 50°C and 1 atm.

Fundamentals of Engineering (FE) Exam Problems

6-116 The transition from laminar flow to turbulent flow in a forced convection situation is determined by which one of the following dimensionless numbers?

- Grasshof
- Nusselt
- Reynolds
- Stanton
- Mach

6-117 The _____ number is a significant dimensionless parameter for forced convection, and the _____ number is a significant dimensionless parameter for natural convection.

- Reynolds, Grasshof
- Reynolds, Mach
- Reynolds, Eckert
- Reynolds, Schmidt
- Grashof, Sherwood

6-118 In any forced or natural convection situation, the velocity of the flowing fluid is zero where the fluid wets any stationary surface. The magnitude of heat flux where the fluid wets a stationary surface is given by

- $k(T_{\text{fluid}} - T_{\text{wall}})$
- $k \frac{dT}{dy} \Big|_{\text{wall}}$
- $k \frac{d^2T}{dy^2} \Big|_{\text{wall}}$
- $h \frac{dT}{dy} \Big|_{\text{wall}}$
- None of them

6-119 The coefficient of friction C_f for a fluid flowing across a surface in terms of the surface shear stress, τ_w , is given by

- $2\rho V^2/\tau_w$
- $2\tau_w/\rho V^2$
- $2\tau_w/\rho V^2 \Delta T$
- $4\tau_w/\rho V^2$
- None of them

6-120 Most correlations for the convection heat transfer coefficient use the dimensionless Nusselt number, which is defined as

- (a) h/k (b) k/h (c) hL_c/k
 (d) kL_c/h (e) $k/\rho c_p$

6-121 For the same initial conditions, one can expect the laminar thermal and momentum boundary layers on a flat plate to have the same thickness when the Prandtl number of the flowing fluid is

- (a) Close to zero (b) Small (c) Approximately one
 (d) Large (e) Very large

6-122 One can expect the heat transfer coefficient for turbulent flow to be _____ for laminar flow.

- (a) less than (b) same as (c) greater than

6-123 An electrical water ($k = 0.61 \text{ W/m}\cdot\text{K}$) heater uses natural convection to transfer heat from a 1-cm-diameter by 0.65-m-long, 110 V electrical resistance heater to the water. During operation, the surface temperature of this heater is 120°C while the temperature of the water is 35°C , and the Nusselt number (based on the diameter) is 6. Considering only

the side surface of the heater (and thus $A = \pi DL$), the current passing through the electrical heating element is

- (a) 3.2 A (b) 3.7 A (c) 4.6 A (d) 5.8 A (e) 6.6 A

6-124 In turbulent flow, one can estimate the Nusselt number using the analogy between heat and momentum transfer (Colburn analogy). This analogy relates the Nusselt number to the coefficient of friction, C_f , as

- (a) $\text{Nu} = 0.5 C_f \text{Re} \text{Pr}^{1/3}$ (b) $\text{Nu} = 0.5 C_f \text{Re} \text{Pr}^{2/3}$
 (c) $\text{Nu} = C_f \text{Re} \text{Pr}^{1/3}$ (d) $\text{Nu} = C_f \text{Re} \text{Pr}^{2/3}$
 (e) $\text{Nu} = C_f \text{Re}^{1/2} \text{Pr}^{1/3}$

Design and Essay Problems

6-125 Design an experiment to measure the viscosity of liquids using a vertical funnel with a cylindrical reservoir of height h and a narrow flow section of diameter D and length L . Making appropriate assumptions, obtain a relation for viscosity in terms of easily measurable quantities such as density and volume flow rate.

6-126 A facility is equipped with a wind tunnel and can measure the friction coefficient for flat surfaces and airfoils. Design an experiment to determine the mean heat transfer coefficient for a surface using friction coefficient data.